

Variance Trading and Market Price of Variance Risk*

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Abstract

This paper develops a new approach for variance trading. We show that the discretely-sampled realized variance can be robustly replicated under very general conditions, including when the price can jump. The replication strategy specifies the exact timing for rebalancing in the underlying. The deviations from the optimal schedule can lead to surprisingly large hedging errors. In the empirical application, we synthesize the prices of the variance contract on S&P 500 index over the period from 01/1990 to 12/2009. We find that the market variance risk is priced, its risk premium is negative and economically very large. The variance risk premium cannot be explained by the known risk factors and option returns.

JEL Classification: G12, G13

Keywords: Variance Risk, Option Valuation, Risk-Neutral Density, Stochastic Volatility

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1 Introduction

Recently, there has been a growing demand for new instruments to hedge shifts in volatility. Several innovative derivative instruments have been introduced in over-the-counter market. One important example is variance swaps. A variance swap has a payoff equal to the difference between the realized variance over a given period and the contract variance. Thus, a variance swap allows traders to bet directly on the level of future variance.¹ Trading in variance swaps has grown dramatically in the aftermath of the recent financial turmoils, such as the summer of 1997, the fall of 1998, the fall of 2001, and the fall of 2008. In May 2004, CBOE launched a closely related instrument, the futures on the S&P 500 realized variance.

This paper proposes a novel approach for pricing of variance instruments without the assumption of complete markets. To explain our approach, suppose first that an underlying asset follows a continuous price process F_t . Let $IV_T = IV(0, T)$ denote the continuously-sampled integrated variance over $[0, T]$ and consider the variance contract, which at maturity T has a payoff of IV_T . The valuation approach for this contract has been developed by pathbreaking work of Dupire (1993), Neuberger (1994), Carr and Madan (1998), Demeterfi, Derman, Kamal, and Zou (1999), and Britten-Jones and Neuberger (2000). The key insight in these papers is that the payoff IV_T can be robustly replicated by a static portfolio of standard options and dynamic trading in the underlying asset. The existing approach is very elegant, but has several important limitations. First, it requires the price process F_t to be continuous. This can be restrictive in view of ample evidence of jumps in many financial assets. Second, the approach requires continuous trading in the underlying asset, which is infeasible in practice. Finally and most importantly, the existing approach focuses on valuing the *integrated* variance IV_T , which is unobservable in practice. The actually traded instruments, such as OTC variance swaps or the CBOE futures, are all based on the discretely-sampled *realized* variance, typically computed from daily returns.

Specifically, let $RV_T = RV(\mathcal{T})$ denote the realized variance computed using the sampling partition $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$ with equidistant trading dates $0 = t_0 < t_1 < \dots < t_n = T$, and let $\Delta t = t_i - t_{i-1}$. In the limit $\Delta t \rightarrow 0$, the realized variance RV_T converges to the integrated variance IV_T , however, the difference between the two can be very large in typical situations. For example, consider a trader who needs to replicate the payoff RV_T with $T = 1$ month and $\Delta t = 1$ day. If the trader follows the existing replication strategy for the payoff IV_T , then the average hedging error is close to 40%. In other words, the existing approach exposes the trader to huge risks and would be unacceptable in practice.

The first contribution of our paper is to develop a novel approach for valuing the payoffs based on the *empirically relevant* realized variance, as opposed to the *theoretical* integrated variance. Our main theoretical result demonstrates that the exact replication of the discretely-sampled realized variance RV_T is possible under very general conditions.² The market does not have to be complete. The underlying price process can be completely general and, in particular, it can have jumps. There is no assumption on the size or frequency of jumps. The partition $\mathcal{T}$ can be arbitrary: Δt does not have to be small and the partition does not have to be equidistant. Our approach nests the existing approach as a special case when (1) $\Delta t \rightarrow 0$ and (2) there are no jumps. The replication

¹Both *variance* and *volatility* swaps are available over-the-counter, with volatility defined as the square root of the variance. Although the two types of swaps are closely related, variance swaps appear to be more popular, due to the fact that variance swaps offer useful theoretical advantages and as such are easier to hedge.

²In this paper, we study three variations of the realized variance. The three versions coincide in the continuous-time limit with no jumps. Although only one version allows for robust replication, we show that the differences between the three versions are small for most practical purposes. In contrast, the difference between the integrated variance and any versions of the realized variance is typically very large.

strategy for the realized variance RV_T consists of two components: (i) a static portfolio of standard puts and calls with the same time to maturity T and different strikes, and (ii) a dynamic trading strategy in the underlying asset on dates $t_0 < t_1 < \dots < t_n$.³

Suppose that the realized variance is computed from daily closing prices. To replicate the variance contract, the trader must at time-0 form a specific option portfolio and then trade in the underlying asset once a day, at the market close. The replication strategy specifies the *exact* timing for rebalancing and we demonstrate that if the trader deviates from the optimal timing even slightly, the hedging errors could be surprisingly high. For example, if the trader rebalances too frequently (twice a day) or not frequently enough (every two days), then the average error of the replication strategy is around 26-29%. Furthermore, when the trader rebalances once a day, but not necessarily at the market close, the hedging errors could again be surprisingly large. For example, if the trader rebalances only 5 minutes prior to the market close, the average hedging error is about 7%. This finding has important implications. As trading in the variance products becomes more active, there might be a predictable price impact at the market close due to the fact that traders, trying to minimize their hedging risks, will execute their orders at essentially the same time. For related reasons, one might also be concerned about possible price manipulation around the market close.

The second contribution of our paper is to introduce the new class of so-called *generalized* variance contracts, which could be robustly replicated under the same general conditions. In a generalized variance contract, the exposure to the variance can vary over time according to some weight function $\omega(F_t)$. The weight function allows one to construct flexible, custom-tailored contracts, where the exposure to the variance is adjusted in response to market conditions. For example, the choice of $\omega(F) = F^a$ leads to the class of the power-price weighted variance contracts (PPWVC). When $a > 0$, the variance exposure increases with the price level F_t . This property might be appealing to traders due to the fact that for many financial assets, the price and the variance are negatively correlated. Other important examples of generalized variance contracts include the barrier and corridor variance contracts, see, for example, Andersen and Bondarenko (2007, 2010) and Andersen, Bondarenko, and Gonzalez-Perez (2011). We extend our main theoretical result to the whole class of generalized variance contracts and derive the corresponding replication strategy.

The third contribution of our paper is to document the economic properties of the variance risk. Our theoretical result implies that, even when the variance contract is not traded, its unique arbitrage-free price can be inferred from prices of standard put and call options. In the empirical application, we synthesize the prices of the variance contract on the S&P 500 index and generate a series of monthly variance return r_v . Our sample is from 01/1990 to 12/2009 and allows us to study historical performance of the variance contract over a very long period. The main results can be summarized as follows:

- Consistent with other studies, we find that the variance risk is priced, its risk premium is negative and very large in economic terms. In particular, the unconditional variance risk premium is estimated as -26.30% per month. Intuitively, because the variance contract pays off when the realized variance is high, investors are willing to pay high prices for the hedge against sharp increases in the market variance.
- The variance return r_v and the market return r_m are negatively correlated. This raises an important question as to whether the variance risk is priced *beyond* its correlation with the

³The existing approach consists of the same components (i) and (ii), except the underlying must be rebalanced continuously.

market return. We find that the negative correlation accounts for only a small portion of the variance risk premium (-2.72% per month). The remaining portion (-23.58% per month) can be interpreted as the compensation for the “pure” variance risk.

- Similarly, the variance risk premium cannot be explained by other known risk factors, such as the size factor, the value factor, the momentum factor, as well as returns on ATM and OTM puts and calls.
- Over the 20-year period, selling the variance contract would have resulted in very high profits over the studied 20-year period. This strategy achieves the risk-adjusted performance measures, such the Sharpe ratio, Stutzer index, and Sortino ratio, which are 4-5 times higher than those for the market itself and 1.6-1.8 times as large as for selling the ATM put option.

Overall, we conclude that the variance represents a new risk factor, which the market prices very heavily. This has important implications for explaining returns on assets which are sensitive to changes in the market volatility. Even a slight exposure to the variance risk could have a non-trivial impact, because the variance risk premium is so large.

The remainder of the paper is organized as follows. This section concludes with the review of the related literature. Section 2 presents the new methodology for valuing the variance contract. Section 3 applies the methodology on S&P 500 futures options and documents the statistical properties of the variance return. Section 4 concludes.

1.1 Related Literature

The original approach for replicating the variance payoff has been developed in Dupire (1993), Neuberger (1994), Carr and Madan (1998), Demeterfi, Derman, Kamal, and Zou (1999), Britten-Jones and Neuberger (2000). More recently, Carr and Lee (2009) develop valuation of arbitrary functions of the variance, although their approach requires continuous trading in options. Jiang and Tian (2005) and Carr and Wu (2009) investigate the effect of jumps on variance trading. All these papers focus on the integrated variance and assume that continuous monitoring and rebalancing are possible.

The issue whether the variance risk is priced has been investigated in a number papers. Coval and Shumway (2001) study daily returns on the ATM zero-beta straddles and argue that the variance risk premium is negative. In their approach, some (weak) parametric assumptions on the option pricing model are needed (to compute option betas), and the sensitivity of the ATM zero-beta straddle to the variance risk is model-specific. Furthermore, the approach requires frequent trading in options, which would be prohibitively costly in practice.

Bakshi and Kapadia (2003a, 2003b) investigate statistical properties of delta-hedged gains of equity call options. In the stochastic volatility framework, they show that the sign of delta-hedged option portfolios corresponds to the sign of the variance risk premium. Empirically, they find that the variance risk is priced and is negative. In their approach, some (weak) parametric assumptions are required (to compute delta-hedges), and the magnitude of the variance risk premium cannot be determined.

There is also an extensive strand of the literature, in which structural models, such as Hull and White (1987), Heston (1993), Bates (2000), are calibrated to option data. Recent examples include Bakshi, Cao, and Chen (1997), Guo (1998), Bates (2000), Chernov and Ghysels (2000), and Pan (2002), Broadie, Chernov, and Johannes (2007), among others. These approaches make the strongest assumptions and are subject to model misspecification.

More recently, Ang, Hodrick, Xing, and Zhang (2008) investigate performance of stock portfolios ranked by their sensitivity to changes in the VIX index. They find evidence of negative volatility risk premium in the cross-section of stock returns. Todorov (2010), Bollerslev, Gibson, and Zhou (2011), and Wu (2011) use high-frequency returns and the CBOE VIX index to study the dynamics of the variance risk premium. Egloff, Leippold, and Wu (2010) estimate a two-factor affine model to explain the term structure variation of OTC variance swap rates.

2 Theoretical Methodology

We focus on a frictionless and arbitrage-free market with a single risky asset over the period $[0, T]$. We consider various derivative instruments on the risky asset which all pay out at date T . To avoid irrelevant complications with interest rates and dividends, we work throughout with *forward* prices. Thus, all prices in this section are forward prices for settlement at time T .⁴ Formally, let F_t denote an adapted process on a standard filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t \in [0, T]}, \mathbb{P})$, where F_t represents the forward price of the underlying asset. In this setting, there exists a risk-neutral (or, equivalent martingale) measure $\mathbb{Q}$, for which the expected return of all traded securities are martingales with respect to $(\Omega, \mathcal{F}, (\mathcal{F})_{t \in [0, T]}, \mathbb{Q})$.

Let $P_t(K) = P(K, T; F_t, t)$ and $C_t(K) = C(K, T; F_t, t)$ denote the forward prices of European put and call with strike K and maturity T . The option prices can be computed using the risk-neutral density (RND):

$$C_t(K) = \int_0^\infty (F_T - K)^+ h_t(F_T) dF_T, \quad P_t(K) = \int_0^\infty (K - F_T)^+ h_t(F_T) dF_T,$$

where $h_t(F_T) = h(F_T, T; F_t, t)$ is RND. RND satisfies the relationship first discovered in Ross (1976), Breeden and Litzenberger (1978), Banz and Miller (1978):

$$h_t(F_T) = \frac{\partial^2 C_t(K)}{\partial K^2} \Big|_{K=F_T} = \frac{\partial^2 P_t(K)}{\partial K^2} \Big|_{K=F_T}. \quad (1)$$

For future needs, it is convenient to denote as $M_t(K) = \min(P_t(K), C_t(K))$ the minimum of the put and call. In other words, of the two vanilla options with strike K , $M_t(K)$ equals the price of the one that is currently out-of-the-money.

2.1 The Integrated Variance Contract

To present the main idea of our approach, we initially assume that F_t follows a continuous process. Specifically, suppose that under the risk-neutral measure F_t evolves as

$$\frac{dF_t}{F_t} = \sqrt{v_t} dB_t^*, \quad (2)$$

where B_t^* is a standard Brownian motion and v_t is a strictly positive adapted process. The instantaneous variance v_t can be a very general stochastic process. In particular, it can have jumps. It is only assumed that the following integral exists:

$$IV(0, T) := \int_0^T \left(\frac{dF_t}{F_t} \right)^2 = \int_0^T v_t dt,$$

⁴In the empirical application, we approximate the forward price F_t with the S&P 500 futures.

where $IV(0, T)$ denotes the integrated variance over the period $[0, T]$.

Consider the *variance contract*, which at maturity T has the payoff of $IV_T = IV(0, T)$. We are interested in the time-0 value of this derivative security, $E_0^*[IV_T]$, where $E^*[\cdot]$ denotes the expectation under the risk-neutral measure. A comment on the terminology might be in order. In practice, the variance is most commonly traded using variance swaps. The variance swap has a value of zero at inception and at maturity it pays the difference between IV_T and the fixed leg $E_0^*[IV_T]$. The variance contract can be obtained by entering the variance swap and investing the fixed leg of the swap at the risk-free rate. Focusing on the variance contract, as opposed to the variance swap, allows for an easier comparison with other securities, such as put and call options, and for a natural definition of the return on the variance.

The valuation approach for the variance contract has been developed in path-breaking work of Dupire (1993) and Neuberger (1994).⁵ The key role in their approach is played by the following portfolio of put and call options:

$$OP_T := 2 \left(\int_0^{F_0} \frac{P_T(K)}{K^2} dK + \int_{F_0}^{\infty} \frac{C_T(K)}{K^2} dK \right) = 2 \int_0^{\infty} \frac{M_T(K)}{K^2} dK. \quad (3)$$

Portfolio OP_T consists of European puts and calls with the same maturity T and a continuum of strikes K from 0 to ∞ , where portfolio weights are equal to $2/K^2$. Puts are used for low strikes ($K \leq F_0$) and calls are used for high strikes ($K \geq F_0$), which reflects the fact that out-of-the-money (OTM) options are more liquid than in-the-money (ITM) options. The value of this portfolio at time-0 is denoted as

$$U_0 := E_0^*[OP_T] = 2 \int_0^{\infty} \frac{M_0(K)}{K^2} dK. \quad (4)$$

Proposition 1 establishes the replication strategy for the variance contract. For completeness, its proof is provided in Appendix A.

Proposition 1 *If the process F_t is continuous, then the payoff IV_T can be replicated by*

(i) *a time- T payoff equal to OP_T ;*

(ii) *a dynamic strategy, which at any time t holds $2 \left(\frac{1}{F_t} - \frac{1}{F_0} \right)$ shares of the underlying.*

Therefore,

$$E_0^*[IV_T] = U_0.$$

Proof: See Appendix A.

Proposition 1 states that the two payoffs, IV_T and OP_T , have the same market values at time-0. This does *not* mean that the two payoffs are the same at maturity T . In fact, the two payoffs are quite different, because IV_T depends on the whole path of F_t over the period $[0, T]$, while OP_T only depends on the final value of the underlying at maturity T . However, the difference between the path-dependent payoff IV_T and path-independent payoff OP_T can be perfectly replicated by continuous trading in the underlying, with the risk-neutral expectation of the dynamic trading strategy (ii) being zero.

In view of Proposition 1, one can estimate the market price of the variance contract U_0 in the following way. Suppose one observes market prices of standard calls and puts with various strikes and the same maturity T . One can interpolate option prices with available strikes to obtain the call and put pricing functions $C_0(K)$ and $P_0(K)$ for a continuum of strikes. The two integrals in (4) are then evaluated by integrating $C_0(K)$ and $P_0(K)$ over K with the weight function $1/K^2$.

⁵See also Carr and Madan (1998), Demeterfi *et al* (1999), Britten-Jones and Neuberger (2000).

2.2 The Realized Variance Contract

The approach from the previous subsection is very elegant, but it has important limitations. First, although the approach imposes little structure on the variance process v_t , it does require the price process F_t to be continuous. This assumption can be restrictive: by now the empirical literature has documented strong evidence of jumps in many financial assets. Second, the approach requires continuous trading in the underlying asset, which is infeasible in practice. Even though futures can be traded with low transaction costs, the overall costs of continuous rebalancing would be infinite. Finally and most importantly, the existing approach focuses on valuing the *integrated* variance IV_T , which is unobservable. The actually traded instruments are all based on the discretely-sampled *realized* variance, typically computed from low-frequency returns.

We now propose a novel way around these limitations. From now on, we no longer require that the process F_t is continuous. Let $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$ denote a sampling partition with trading dates $0 = t_0 < t_1 < \dots < t_n = T$ and let $\Delta t := \max_i(t_i - t_{i-1})$. Oftentimes, the trading dates are set at equal intervals, in which case $t_i - t_{i-1} = \Delta t = T/n$ for all i , but, in general, this does not have to be the case. Let $x_i = F_i/F_{i-1} - 1$ denote the return over $[t_{i-1}, t_i]$ and where for simplicity we denote $F_i = F_{t_i}$. The realized variance at the partition $\mathcal{T}$ can be defined in several ways, including

$$RV_T^{(1)} = RV^{(1)}(\mathcal{T}) := \sum_{i=1}^n [\ln(1 + x_i)]^2, \quad (5)$$

$$RV_T^{(2)} = RV^{(2)}(\mathcal{T}) := \sum_{i=1}^n x_i^2. \quad (6)$$

Over-the-counter variance swaps all use the daily sampling. Most of them are based on definition (5), but some are based on definition (6). The CBOE futures on the realized variance also use daily sampling and are based on definition (5).

We also introduce another specification for the realized variance:

$$RV_T^{(3)} = RV^{(3)}(\mathcal{T}) := 2 \sum_{i=1}^n (x_i - \ln(1 + x_i)). \quad (7)$$

Although the specification in (7) may not “look” like variance, it has all the intuitive properties of variance. In particular,

- $RV_T^{(3)}$ is always nonnegative;
- In the special case when the price process F_t is continuous, all three measures $RV_T^{(1)}$, $RV_T^{(2)}$, and $RV_T^{(3)}$ converge to IV_T as Δt approaches zero;
- In typical applications, $RV_T^{(1)}$, $RV_T^{(2)}$, and $RV_T^{(3)}$ are very close to each other.

Figure 1 compares the three functions

$$\begin{aligned} f^{(1)}(x) &= [\ln(1 + x)]^2, \\ f^{(2)}(x) &= x^2, \\ f^{(3)}(x) &= 2(x - \ln(1 + x)), \end{aligned}$$

used in the definitions of $RV_T^{(1)}$, $RV_T^{(2)}$, and $RV_T^{(3)}$. Several points are worth mentioning. First, the function $f^{(3)}(x)$ is positive for all returns $|x| > 0$. Second, the three functions are very close to

each other for the range of typical daily returns. In particular, the differences between them are very small when $|x| \leq 10\%$. Third, the function $f^{(3)}(x)$ always lies between $f^{(1)}(x)$ and $f^{(2)}(x)$. Finally, the function $f^{(3)}(x)$ is closer to $f^{(1)}(x)$ than to $f^{(2)}(x)$. The following proposition explains what is so special about the new definition in (7).

Proposition 2 *For any partition $\mathcal{T}$, the payoff $RV_T^{(3)}$ can be perfectly replicated by*

(i) *a time- T payoff equal to OP_T ;*

(ii) *a dynamic trading strategy, which is rebalanced on dates $t_i \in \mathcal{T}$ to maintain $2\left(\frac{1}{F_i} - \frac{1}{F_0}\right)$ shares of the underlying.*

Therefore,

$$E_0^* \left[RV_T^{(3)} \right] = U_0.$$

Proof: See Appendix A.

The replication strategy for the realized variance $RV_T^{(3)}$ parallels that of Proposition 1. It consists of (i) the same path-independent payoff OP_T , and (ii) the modified dynamic trading strategy, where the underlying is rebalanced not continuously but on discrete dates. Proposition 1 may be viewed as a special case of Proposition 2 when there are no jumps and $\Delta t \rightarrow 0$. To formalize this intuition, Appendix B examines the three definitions of the realized variance in the continuous-time limit.

Using Taylor' expansion, it is easy to verify that

$$\begin{aligned} f^{(1)}(x) &= 2(x - \ln(1+x)) - \frac{1}{3}x^3 + O(x^4) = f^{(3)}(x) - \frac{1}{3}x^3 + O(x^4), \\ f^{(2)}(x) &= 2(x - \ln(1+x)) + \frac{2}{3}x^3 + O(x^4) = f^{(3)}(x) + \frac{2}{3}x^3 + O(x^4). \end{aligned}$$

This implies that

$$\begin{aligned} E_0^* \left[RV_T^{(1)} \right] &= U_0 - \frac{1}{3} \sum_{i=1}^n E_0^* \left[x_i^3 \right] + \sum_{i=1}^n E_0^* \left[O(x_i^4) \right], \\ E_0^* \left[RV_T^{(2)} \right] &= U_0 + \frac{2}{3} \sum_{i=1}^n E_0^* \left[x_i^3 \right] + \sum_{i=1}^n E_0^* \left[O(x_i^4) \right]. \end{aligned}$$

The payoff $RV_T^{(1)}$ has a negative (unhedged) exposure to cubed returns, while the payoff $RV_T^{(2)}$ has a positive exposure to cubed returns with twice the magnitude.⁶ If under the risk-neutral measure negative jumps on average dominate positive jumps, then U_0 understates the market price of $RV_T^{(1)}$ but overstates it for $RV_T^{(2)}$.

While its proof is very simple, Proposition 2 has important implications. First, the result holds for completely general price processes, not only continuous ones. The only maintained assumption is the absence of arbitrage. Second, the result holds for any sampling partition $\mathcal{T}$. In particular, $\mathcal{T}$ does not have to be equidistant and Δt does not have to be small. Third, the replication strategy for $RV_T^{(3)}$ requires *discrete* rebalancing in the underlying and the proposition identifies the exact timing for rebalancing. Below we discuss some of the implications of Proposition 2 in more detail.

⁶Clearly, the three versions of the realized variance are related as $RV_T^{(3)} \approx \frac{2}{3}RV_T^{(1)} + \frac{1}{3}RV_T^{(2)}$.

2.2.1 Allowing for jumps

As mentioned earlier, an important limitation of the existing approach is its reliance on the assumption of continuous price process. The empirical literature have presented strong evidence that prices of many financial assets can jump. Recent studies for the S&P 500 index include Bakshi, Cao, and Chen (1997), Andersen, Benzoni, and Lund (2002), Pan (2002), Eraker, Johannes, and Polson (2003), Aït-Sahalia (2002), Carr and Wu (2003), among others.

Allowing for jumps is important for many derivatives, but this is especially the case with the variance contract, whose payoff could be very sensitive to extreme (but, perhaps, rare) returns. To see this intuitively, suppose that, on a typical day, the volatility of the underlying return is 1%, which corresponds to about 16% annually. Consider also a day when the underlying jumps by 10%. The contribution of this extreme day to the variance payoff will be equivalent to 100 days with no jumps, or about 5 months. Similarly, a single day when the underlying jumps by 30% will be equivalent to 900 “normal” days, or more than 3.5 years.

2.2.2 Traded instruments are based on *realized* variance

The existing literature focuses almost exclusively on valuing payoffs based on the integrated variance, which is an unobservable, theoretical quantity. On the other hand, traded instruments are based on the *realized* variance, typically computed from daily returns. The implicit presumption in the literature is that the integrated variance and the realized variance are close substitutes and that the difference between two can be safely ignored. However, while it is true that the realized variance converges to the integrated variance in the limit $\Delta t \rightarrow 0$, the difference between the two can be very large in typical applications. Stated differently, the existing literature develops the replication strategy for the “wrong” payoff.

To clarify this point, suppose that $T = 1$ month and $\mathcal{T}$ is the equidistant sampling partition with $\Delta t = 1$ day. Consider a trader who has sold the variance contract with the payoff based on the realized variance RV_T (which could be computed according to any of the three definitions $RV_T^{(1)}$, $RV_T^{(2)}$, or $RV_T^{(3)}$). The trader wishes to dynamically replicate the payoff by following the replication strategy of Proposition 1. In the absence of jumps and transaction costs, the trader is able to perfectly replicate the payoff IV_T . The problem, however, is that the target payoff RV_T will typically differ from the replicated payoff IV_T considerably.

To gauge the accuracy of the replicating strategy, we consider three canonical models: Black-Scholes (1973, BS), Merton (1976) jump-diffusion (JD), and Heston (1993) stochastic volatility (SV). The parameters for the models are reported in Table 1. For each model, we conduct a Monte Carlo experiment and compute the hedging error (HE) as RMSE of the relative error. Table 2 reports the results for the three models and three definitions of the realized variance. In all cases, HE is around 40%, suggesting that the considered replication strategy exposes the trader to huge risks and would be completely unacceptable in practice. The results are very similar across the models and are virtually indistinguishable across the three definitions of the realized variance $RV_T^{(1)}$, $RV_T^{(2)}$, or $RV_T^{(3)}$.

The important message of this subsection is that the replication strategy in Proposition 1, designed to replicate the *theoretical* continuously-sampled variance, is simply inappropriate for replicating the *empirically relevant* payoffs sampled at the daily frequency.

2.2.3 Optimal timing for rebalancing

Proposition 2 establishes an interesting fact that, in order to perfectly replicate the payoff $RV_T^{(3)}$, one must rebalance the position in the underlying on a finite number of fixed dates. If the realized

variance is computed using the sampling partition $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$, then rebalancing must take place on the same dates $t_0, t_1, \dots, t_n$.

On surface, this fact might appear unremarkable. After all, the path-dependent payoff $RV_T^{(3)}$ is only affected by the values of the underlying on the sampling dates $t_0, t_1, \dots, t_n$ and not by what happens to the underlying between them. Is it then surprising that the sampling dates play a special role in the replication strategy? However, this property is far from obvious. One can recall that vanilla European put and call options have payoffs which only depend on the value of the underlying at maturity, F_T . Yet, to replicate these payoffs, continuous trading in the underlying is necessary. Similarly, exotic path-dependent options (such as lookback or Asian options) are often defined using a finite number of fixing dates, but they also require continuous rebalancing.

Table 3 reports on two experiments which illustrate that, if the trader deviates from the optimal rebalancing strategy even slightly, the hedging errors could be surprisingly high. Let $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$ be an equidistant sampling partition with $\Delta t = 1$ day. This partition is used to compute the realized variance $RV_T = RV(\mathcal{T})$. To replicate this payoff, the trader follows the strategy described in Proposition 2, except the underlying is rebalanced using a different partition $\mathcal{T}' = \{t'_0, t'_1, \dots, t'_m\}$, where for simplicity we assume that $t'_0 = t_0$ and $t'_m = t_n = T$. This replication strategy generates payoff RV_T' , which is generally different from the target payoff RV_T . In addition to the parametric models BS, JD, and SV, we also compute hedging errors for the S&P 500 futures based on real data.

In the first experiment, we assume that the partition $\mathcal{T}'$ is equidistant and its sampling interval $\Delta t' = 1$ min, 10 min, 1 hour, 1/2 day, 1 day, 2 days, and 5 days. Panel A shows that rebalancing too frequently or too infrequently leads to considerable hedging errors. For example, when $\Delta t' = 1/2$ day (2 days), the average error of the replication strategy is about 27% (29%) for the three models. For the S&P 500, the average error of the replication strategy is about 25% (27%) when $\Delta t' = 1/2$ day (2 days).

In the second experiment, we assume that the trader rebalances once a day, but not necessarily at the exact times specified by the partition $\mathcal{T}$. Specifically, although traded instruments typically use daily closing prices, suppose instead that the trader rebalances at times $\mathcal{T}' = \{t_0, t'_1, \dots, t'_{n-1}, t_n\}$, where $t'_i = t_i - \delta t$, $i = 1, \dots, n-1$. Panel B reports the results for $\delta t = 1, 5, 15$, and 60 minutes. The striking finding in the table is that the hedging errors are very sensitive to even slightest non-synchronicity. If trades are executed only 1 (5) minutes before the market close, the average error of the replication strategy is non-negligible: 3.1% (7.0%) for BS, 3.1% (6.8%) for JD, 3.2% (7.1%) for SV, and 4.9% (7.5%) for the S&P 500. In other words, deviating even slightly from the optimal rebalancing schedule would expose the trader to considerable risk.⁷ This observation has important implications. As trading in the variance products becomes more active, there might be a predictable price impact at the market close due to the fact that traders, seeking to minimize their hedging risks, will execute their orders at essentially the same time. For related reasons, one might also be concerned about possible price manipulation around the market close. In particular, if the closing price of the underlying is pushed up (down) on days with large positive (negative) returns, then the payoff of the variance contract can be increased by a sizable amount. This issue becomes particularly important for the variance contracts on less

⁷Broadie and Jain (2008) investigate the effect of discrete sampling and jumps on pricing of variance swaps. They consider several parametric models and show that the effect of discrete sampling on the fair price of the variance swap is typically small while the effect of jumps can be significant. This is consistent with Proposition 2, which states that the market price of the realized variance $E_0^*[RV_T^{(3)}]$ does not depend on Δt at all. Similarly, the market prices of the realized variances $RV_T^{(1)}$ and $RV_T^{(2)}$ are not very sensitive to Δt . However, for *hedging* purposes, the effect of discrete sampling cannot be ignored, as the payoffs $RV_T^{(j)}$ vary considerably with Δt . On the other hand, jumps are important for both *pricing* and *hedging*.

liquid individual stocks.

2.3 The Generalized Variance Contracts

In this subsection, we introduce the *generalized* variance contracts, which can be robustly replicated under the same general conditions as before. Let $g(F)$ be a general function which is twice differentiable almost everywhere. For given partition $\mathcal{T} = \{t_0, t_1, \dots, t_n\}$ and function $g(F)$, we can consider three versions of the generalized realized variance:

$$RV_T^{(1)}(g) = RV^{(1)}(g, \mathcal{T}) := \sum_{i=1}^n g''(F_{i-1}) F_{i-1}^2 \left[\ln \frac{F_i}{F_{i-1}} \right]^2, \quad (8)$$

$$RV_T^{(2)}(g) = RV^{(2)}(g, \mathcal{T}) := \sum_{i=1}^n g''(F_{i-1}) (F_i - F_{i-1})^2, \quad (9)$$

$$RV_T^{(3)}(g) = RV^{(3)}(g, \mathcal{T}) := 2 \sum_{i=1}^n [g(F_i) - g(F_{i-1}) - g'(F_{i-1})(F_i - F_{i-1})]. \quad (10)$$

The three measures only differ in terms of order $O(\Delta F_i^3)$ and, typically, are very close to each other.⁸ Appendix B looks at the continuous time limit of the three measures. In particular, it shows that, if the asset price is governed by a pure diffusion (2), then in the limit $\Delta t \rightarrow 0$ the three definitions all converge to the *generalized* integrated variance $IV_T(g) = \int_0^T g''(F_t) F_t^2 v_t dt$. In the latter, the variance is integrated with the *weight function* $\omega(F) := g''(F) F^2$, which depends on the price level. Subsection 2.3.2 discusses several important choices for $\omega(F)$.⁹

Of the three measures, only the payoff $RV_T^{(3)}(g)$ admits model-independent replication. Before deriving the general result in the next subsection, it is convenient to introduce additional notation. First, we define the following payoff at maturity T :

$$OP_T(g) := 2 \int_0^\infty g''(K) M_T(K) dK.$$

As before, the path-independent payoff $OP_T(g)$ can be statically replicated by a portfolio of European options with maturity T and a continuum of strikes K from 0 to ∞ , where portfolio weights are equal to $2g''(K)$. Second, we define a new function:

$$G(x_1, x_2) = G(g, x_1, x_2) := 2 [g(x_2) - g(x_1) - g'(x_1)(x_2 - x_1)]. \quad (11)$$

G depends on specific choice of g , but, for notational convenience, we will suppress this explicit dependence. With this notation, the definition in (10) simplifies to $RV_T^{(3)}(g) = \sum_{i=1}^n G(F_{i-1}, F_i)$.

⁸It is easy to verify that

$$\begin{aligned} RV_T^{(1)}(g) &= RV_T^{(3)}(g) + \sum_{i=1}^n \left(g''(F_{i-1}) F_{i-1}^2 - \frac{1}{3} g'''(F_{i-1}) F_{i-1}^3 \right) x_i^3 + \sum_{i=1}^n O(x_i^4), \\ RV_T^{(2)}(g) &= RV_T^{(3)}(g) + \sum_{i=1}^n \left(-\frac{1}{3} g'''(F_{i-1}) F_{i-1}^3 \right) x_i^3 + \sum_{i=1}^n O(x_i^4), \end{aligned}$$

implying that the (unhedged) exposure to cubed returns is determined by function $(g''(x)x^2 - \frac{1}{3}g'''(x)x^3)$ for $RV_T^{(1)}(g)$ and by function $-\frac{1}{3}g'''(x)x^3$ for $RV_T^{(2)}(g)$.

⁹See also Lee (2010), who extends the existing model-free approach to weighted variance swaps.

Note also that the second derivative $g''(x)$ is typically positive in applications. Under this additional assumption the weight function $\omega(x)$ is positive too and $G(x, y)$ exhibits the basic properties of the “variance”. Indeed, for small Δx ,

$$G(x, x + \Delta x) = G(x, x) + G_2(x, x)\Delta x + G_{22}(x, x)\frac{\Delta x^2}{2} + O(\Delta x^3) = g''(x)\Delta x^2 + O(\Delta x^3).$$

Thus, there are no zero and first order terms and the leading term is proportional to the squared return.

2.3.1 The Aggregation Property

In an important recent paper, Neuberger (2012) introduces the class of payoffs which satisfy a certain aggregation property. Let X_t denote an adapted process on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t \in [0, T]}, \mathbb{Q})$. A real-valued function $H(x)$ has the Aggregation Property (AP1) if, for any martingale process X_t and for any times $0 \leq t \leq s \leq u \leq T$,

$$\mathbf{AP1:} \quad E_t^*[H(X_u - X_t)] = E_t^*[H(X_s - X_t)] + E_t^*[H(X_u - X_s)].$$

To appreciate the above definition, suppose that $X_t = F_t$, which is a martingale under the risk-neutral measure. If $H(x)$ satisfies AP1, then the discretely-sampled payoff $\sum_{i=1}^n H(F_i - F_{i-1})$ has the same market price as the path-independent time- T payoff of $H(F_T - F_0)$. Indeed, AP1 and the Law of Iterated Expectations imply that for any partition $\mathcal{T}$,

$$E_0^* \left[\sum_{i=1}^n H(F_i - F_{i-1}) \right] = E_0^* [H(F_T - F_0)],$$

which makes possible the static replication of the path-dependent payoff with standard options. Neuberger studies two main cases: (i) a scalar process $X_t = F_t$, and (ii) a vector process $X_t = (F_t, V_t)$, where the variance process $V_t = \text{Var}_t^*(F_T)$ is assumed to be tradeable. In the first case, he shows that functions $H(x) = x$ and $H(x) = x^2$ are the only ones that satisfy AP1. In the second case, there exist other, non-trivial functions which satisfy AP1.

We now introduce another version of Aggregation Property (AP2)¹⁰, where a function of two variables $H(x, y)$ satisfies the following condition for any martingale X_t and for any times $0 \leq t \leq s \leq u \leq T$:

$$\mathbf{AP2:} \quad E_t^*[H(X_t, X_u)] = E_t^*[H(X_t, X_s)] + E_t^*[H(X_s, X_u)].$$

AP2 is more general than AP1, in the sense that, if $H(x)$ satisfies AP1, then $\tilde{H}(x, y) := H(y - x)$ satisfies AP2. In this paper, we focus only on case (i) of a scalar process $X_t = F_t$ and leave more general case (ii) of a vector process $X_t = (F_t, V_t)$ for future research.

If $H(x, y)$ satisfies AP2, then the discretely-sampled payoff $\sum_{i=1}^n H(F_{i-1}, F_i)$ can again be statically replicated by a time- T payoff of $H(F_0, F_T)$ and

$$E_0^* \left[\sum_{i=1}^n H(F_{i-1}, F_i) \right] = E_0^* [H(F_0, F_T)].$$

The next proposition characterizes the class of functions $H(x, y)$ which satisfy AP2, while Proposition 4 derives the replication strategy for the payoff $\sum_{i=1}^n H(F_{i-1}, F_i)$.

¹⁰I am grateful to an anonymous referee for suggesting to explore this angle.

Proposition 3 $H(x, y)$ satisfies AP2 if and only if there exist univariate functions A and B such that

$$H(x, y) = A(y) - A(x) + B(x)(y - x). \quad (12)$$

Proof: See Appendix A.

Proposition 4 Let $H(x, y) = A(y) - A(x) + B(x)(y - x)$ and $A''(x)$ is continuous almost everywhere. For any partition $\mathcal{T}$, the payoff $\sum_{i=1}^n H(F_{i-1}, F_i)$ can be perfectly replicated by

- (i) a time- T payoff equal to $\int_0^\infty A''(K)M_T(K)dK$;
- (ii) a dynamic trading strategy, which is rebalanced on dates $t_i \in \mathcal{T}$ to maintain $(B(F_i) + A'(F_0))$ shares of the underlying.

Therefore,

$$E_0^* \left[\sum_{i=1}^n H(F_{i-1}, F_i) \right] = \int_0^\infty A''(K)M_0(K)dK.$$

Proof: See Appendix A.

Clearly, function $G(x, y)$ defined in (11) is a special case of (12) with the additional restrictions that $A(x) = 2g(x)$ and $B(x) = -2g'(x)$. For this special case, we obtain the following corollary to Proposition 4.

Corollary 1 For any partition $\mathcal{T}$, the payoff $RV_T^{(3)}(g)$ can be perfectly replicated by

- (i) a time- T payoff equal to $OP_T(g)$;
- (ii) a dynamic trading strategy, which is rebalanced on dates $t_i \in \mathcal{T}$ to maintain $2(g'(F_0) - g'(F_i))$ shares of the underlying.

Therefore,

$$E_0^* [RV_T^{(3)}(g)] = E_0^* [OP_T(g)] = 2 \int_0^\infty g''(K)M_0(K)dK.$$

Corollary 1 establishes the model-free replication strategy for a broad class of payoffs $RV_T^{(3)}(g)$, for completely general price processes and for arbitrary partitions $\mathcal{T}$. Proposition 2 is a special case of Corollary 1, corresponding to the choice of $g(F) = -\ln F$.

2.3.2 Important Examples

One interesting choice of the weight function is $\omega(F) = I(F; B_1, B_2)$, where $I(F; B_1, B_2) := 1[B_1 \leq F \leq B_2]$ is the indicator function. The contract now pays the *corridor* variance, or the variance accumulated only when the price F_t is between the barriers B_1 and B_2 . Carr and Madan (1998) show how to price the corridor variance contract when the process F_t is continuous and under continuous trading.¹¹ Andersen and Bondarenko (2007) study several corridor implied volatility measures constructed for the S&P 500. Interestingly enough, they find that narrow corridors are

¹¹See also Carr and Lewis (2007), who propose an approximate replication strategy for discretely-sampled corridor variance swap.

more useful for predicting future volatility than broad ones. Andersen, Bondarenko, and Gonzalez-Perez (2011) construct a high-frequency corridor volatility index and find it a more accurate and robust alternative to VIX. The special cases of the barriers (B_1, B_2) equal to $(0, F_0)$ and (F_0, ∞) correspond to the down- and up-variance payoffs. Andersen and Bondarenko (2010) study the down and up variances to shed light on pricing of S&P 500 volatility risk.

Another interesting choice for the weight function is $\omega(F) = F^a$, which leads to the class of the power-price weighted variance contracts (PPWVC). The standard variance contract corresponds to the case of $a = 0$. When $a > 0$, the instantaneous exposure to the variance increases with the price level F_t , the property that might be appealing to some traders. For many financial assets, the price and the variance are negatively correlated and the payoff of the PPWVC ($a > 0$) should be less variable than that of the standard variance contract ($a = 0$). Another advantage of the PPWVC with $a > 0$ is that its payoff is defined even when the underlying price can drop to zero, unlike the standard variance contract.¹² The next table summarizes PPWVCs for select values of the power a :

a	$\omega(x) = x^a$	$g''(x)$	$g(x)$	$G(x_1, x_2) = 2[g(x_2) - g(x_1) - g'(x_1)(x_2 - x_1)]$
-1	$\frac{1}{x}$	$\frac{1}{x^3}$	$\frac{1}{2x}$	$(x_2 - x_1)^2 \frac{1}{x_1^2 x_2}$
0	1	$\frac{1}{x^2}$	$-\ln(x)$	$2\left(\frac{x_2}{x_1} - 1 - \ln\left(\frac{x_2}{x_1}\right)\right)$
1	x	$\frac{1}{x}$	$x \ln(x) - x$	$2\left(x_2 \ln\left(\frac{x_2}{x_1}\right) - (x_2 - x_1)\right)$
2	x^2	1	$\frac{1}{2}x^2$	$(x_2 - x_1)^2$
3	x^3	x	$\frac{1}{6}x^3$	$\frac{1}{3}(x_2 - x_1)^2(2x_1 + x_2)$

Although function $G(x_1, x_2)$ generally appears unconventional, as argued above, it possesses the basic properties of the variance. For the case of $a = 2$, however, the payoff of the generalized variance contract takes a very simple and intuitive form:

$$RV_T^{(3)}(g) = \sum_{i=1}^n (F_i - F_{i-1})^2. \quad (13)$$

This contract is closely related to the so-called Simple Variance Swap (SVS) recently proposed by Martin (2013). Martin defines the payoff in terms of the spot price S_t , instead of the forward price F_t . Assuming for simplicity equidistant sampling partition with $\Delta t = t_i - t_{i-1}$, constant risk-free rate r and dividend yield δ , the SVS payoff can be re-written as

$$\sum_{i=1}^n \left(\frac{S_i - S_{i-1}}{S_0 e^{(r-\delta)t_{i-1}}} \right)^2 = \frac{1}{F_0^2} \sum_{i=1}^n \left(F_i e^{(r-\delta)\Delta t} - F_{i-1} \right)^2. \quad (14)$$

Ignoring the inconsequential normalization by $1/F_0^2$, the only difference between the payoffs in (13) and (14) is the factor $e^{(r-\delta)\Delta t}$. Martin shows that the SVS payoff can be robustly replicated in the presence of jumps. His replication strategy becomes exact in the $\Delta t \rightarrow 0$ limit. Martin also

¹²For the case of $a = 1$, PPWVC is sometimes called the gamma contract. In the gamma contract, the exposure to the variance is proportional to the price level F_t . If the market increases by $x\%$, then the variance exposure also increases by $x\%$. The name “gamma” refers to the fact that the gamma contract has a linear cash-gamma, whereas the standard variance contract has a constant cash-gamma. Several banks, including BNP Paribas and Deutsche Bank, have recently started to offer gamma swaps.

derives the analytical bound on the hedging error when $\Delta t > 0$ and shows that it is extremely small independent of the sampling frequency. If the payoff is defined as in (13) (i.e., without the term $e^{(r-\delta)\Delta t}$), then Corollary 1 states that there is exact replication for arbitrary Δt .¹³

Other choices for the weight function $\omega(F)$ can be considered, yielding even more complex, custom-tailored payoffs and allowing investors to bet on even more specific aspects of the variance. For example, one can imagine a contract which pays the corridor power-price-weighted integrated variance. One can also incorporate explicit dependence on time, i.e., $g(F, t)$, but for simplicity we do not pursue such an extension here.

3 Empirical Application

Our data consist of daily prices of the CME options on S&P 500 futures. The risk-free rate is approximated by Treasury bill rates. The sample period is from January 1990 through December 2009. The period is illustrated in Figure 2 which shows the S&P 500 level and the daily return of the S&P 500 futures. Additional information on the dataset is provided in Appendix C.

3.1 Variance Return

The key object of our empirical study is the return on the variance contract where the underlying is the S&P 500 futures. Specifically, let

$$r_v = \frac{RV_T}{E_0^*[RV_T]} - 1 \quad (15)$$

denote the monthly return on the variance contract. The numerator in (15) is the realized variance computed from daily closing prices of the S&P 500 futures. We measure the realized variance according to the definition $RV_T^{(3)}$, in which case exact, model-free pricing is possible and the denominator $E_0^*[RV_T] = U_0$ is known. Figure 3 plots the deviations of the two other realized variances from $RV_T^{(3)}$. Generally, deviations $(RV_T^{(1)} - RV_T^{(3)})$ are positive, consistent with the observation that negative jumps are more prevalent than positive jumps.¹⁴ For most months in our sample, these deviations are small, but they are clearly non-trivial during most turbulent periods. For example, in September of 2008, the deviation $(RV_T^{(1)} - RV_T^{(3)})$ is around 6%, while $(RV_T^{(2)} - RV_T^{(3)})$ is around -12%. Figure 4 provides additional details on this volatile month during the credit crisis. The fact that the three versions of the realized variance diverge the most during volatile periods is unsurprising, but noteworthy. Because the risk-premiums are likely to be particularly elevated during those periods, the *market prices* of the three realized variances could differ substantially. As shown by Broadie and Jain (2008), the market price $E_0^*[RV_T^{(1)}]$ could exceed U_0 considerably if there are jumps and if negative jumps dominate positive ones. For these reasons, we measure the realized variance according to the third definition and, to simplify notation, in what follows we always write $RV_T = RV_T^{(3)}$. The denominator in (15) is equal to U_0 , which in view of Proposition 2 can be inferred from prices of standard S&P 500 options. (Because we compute U_0 from forward prices, the return r_v is already defined in *excess* of the risk-free rate.)

Overall, there are $N = 240$ one-month periods in the sample. Specific steps for calculating the variance return r_v are explained in Appendix C. Briefly, those steps include 1) filtering

¹³Variants of the function $G(x_1, x_2)$ for the special cases of $a = 0, 1$ also appear in Neuberger (2012), who refers to them as the log and the entropy variance contracts.

¹⁴It is also apparent from Figure 3 that deviations $(RV_T^{(2)} - RV_T^{(3)})$ are usually of the opposite sign to and twice as big as deviations $(RV_T^{(1)} - RV_T^{(3)})$. This is consistent with footnote 6.

out unreliable option data, 2) checking that option prices satisfy the theoretical no-arbitrage restrictions, 3) inferring forward prices of European puts and calls, 4) estimating RND, the put and call pricing functions for a continuum of strikes, and 5) estimating the market price of the variance contract using the relationships in (4) and (26).

3.2 The Market Price of the Variance Risk

Table 4 reports various statistics for monthly returns on the variance contract, the S&P 500 futures, and two additional portfolios which will be discussed shortly. We denote as $r_m = F_T/F_0 - 1$ the excess return on the S&P 500 futures contract, which is interpreted as the return on the market portfolio. The average excess return (AER) for the variance contract is -26.30% per month, indicating that the variance risk is priced and its risk premium is negative. Over the same period, AER for the market is about 0.38% per month.

The negative risk premium $\pi_v = E[r_v]$ on the variance contract return is partly due to negative correlation between r_v and r_m . To separate the effect of negative correlation, Panel B of Table 4 reports the Jensen's alpha and beta coefficients with respect to the market return. Specifically, for any return r_i , we compute the alpha and beta coefficients as

$$\alpha_i = E[r_i] - \beta_i E[r_m], \quad \beta_i = \frac{Cov(r_i, r_m)}{Var(r_m)}.$$

The beta coefficient for r_v is -7.22, which accounts for only about -2.72% of the risk-premium π_v ; the alpha coefficient is -23.58% per month. High leverage of the variance contract complicates interpretation of its alpha coefficient. Therefore, Panel B of Table 4 reports several risk-adjusted measures that are unaffected by leverage: Sharpe ratio, Stutzer index, and Sortino ratio. The details of these performance measures are provided in Appendix D. The Sharpe ratio is probably the most popular measure among both academics and practitioners, but it could be misleading when returns are highly non-normal. This is an important consideration for the variance return, which exhibits substantial positive skewness and very high kurtosis. Unlike the Sharpe ratio, the Stutzer index does not assume normality and takes into account the whole shape of the distribution of returns. Specifically, the index penalizes negative skewness and high kurtosis. Importantly, the Stutzer index coincides with the Sharpe ratio when returns are normally distributed. The Sortino ratio differs from the Sharpe ratio in that it does not penalize for large positive returns. Specifically, the risk is now measured by a portfolio's *downside* deviation, instead of standard deviation.

Table 4 demonstrates that selling the variance contract produces a very high Sharpe ratio of 0.42, which is about 5 times as large as the Sharpe ratio for the market. As expected, the Stutzer index for selling the variance contract (0.34) is lower due to non-normality. However, it is still about 4 times higher than the Stutzer index for the market. A similar conclusion obtains on the basis of the Sortino ratio, which measures the risk in terms of losses only. For selling the variance, the Sortino ratio is 0.54, or about 4.5 times larger than for the market.

Table 5 investigates whether the variance risk premium continues to be significantly negative in the presence of other known risk factors. The table reports the results of several OLS specifications where r_v is regressed onto the market return r_m , the Fama-French size and value factors, the momentum factor, as well as returns of four options: ATM and OTM puts ($k = 1.00$ and $k = 0.96$) and ATM and OTM calls ($k = 1.00$ and $k = 1.04$).

In all regressions, the intercept remains highly negative and statistically significant. In particular, in the specification with r_m , SMB, HML, and UMD, all coefficients come out negative. The intuition for this result is as follows. When the market variance unexpectedly increases, the overall market performs poorly in general, however, small stocks, value stocks, and (to a lesser

degree) recent winners tend to do even worse. After accounting for the standard risk factors, the alpha coefficient only slightly declines in absolute value (-20.77%). This suggests that these factors can account for only small portion of the variance risk premium π_v . Similarly, option returns also cannot explain the variance risk premium. In the all-inclusive specification with r_m , SMB, HML, UMD, and the four option returns, the alpha coefficient still remains highly negative (-11.24%) and statistically significant.

Additional insights are provided by Figures 5-6. In Figure 5, the top panel shows the distribution of the variance return over time and indicates the months with largest positive variance returns. The bottom panel contrasts time-series for (1) the market price of the variance contract and (2) the realized variance. For most months, the market price of the variance contract substantially exceeds the subsequently realized variance and the variance return is negative. Occasionally, however, the realized variance spikes up, resulting in a high positive return of the variance contract. Unsurprisingly, the credit crisis of 2008 accounts for the two most dramatic positive variance returns. In Figure 6, the top left panel is the histogram of the variance return r_v , which confirms that the distribution of the variance return is highly non-normal. In particular, the variance return exhibits substantial positive skewness. The top right panel is a scatter plot, which illustrates the negative correlation between returns r_v and r_m (the correlation coefficient is -0.50).

Table 4 also analyzes monthly returns on two special portfolios, the *market-neutral portfolio* (MNP) and the *mean-variance portfolio* (MVP). MNP is a zero-beta portfolio, for which the short position in the variance contract is hedged with the S&P 500 futures to ensure that the sample correlation between the portfolio's return and the market return is zero. Intuitively, MNP accounts for the fact that a portion of the variance return is due to the overall market movements. Formally, the MNP's return is given by $r_{mn} = -r_v + \beta_v r_m$. MVP also consists of the variance contract and the S&P 500 futures. Now, however, the portfolio weights are chosen by the mean-variance optimizer to achieve the highest possible Sharpe ratio. For both MNP and MVP, the Sharpe ratio, the Stutzer ratio, and the Sortino ratio are slightly higher than for the unhedged strategy that sells the variance.

3.3 Comparison to Puts and Calls

One might expect a high correlation between returns on the variance contract and put options. Similar to the variance contract, puts earn considerable negative risk premiums and exhibit positive skewness (see, for example, Bondarenko (2003b).) Still, there are substantial differences in returns on the variance contract and put options.

Table 6 reports various statistics for monthly excess returns of several options on the S&P 500 futures. Specifically, we consider puts and calls with different moneyness as well as the ATM straddle, which is often viewed as a bet on future volatility. We compute returns on *standardized* puts and calls, which at the beginning of each period have constant moneyness $k_0 = K/F_0$ equal to 0.96, 0.98, 1.00, 1.02, and 1.04, and which mature on the same date as the corresponding variance contract. Prices of the options with fixed moneyness and one-month maturity are obtained via interpolation as explained in Appendix C.

Compared to the variance contract, puts generally exhibit a higher degree of leverage and generate more extreme returns. In particular, put returns have much larger standard deviations. As k decreases, put returns become more skewed and have more negative means. As expected, correlations between the variance and put returns are positive, but they are not too extreme, ranging from 0.53 to 0.57. The highest correlation (0.57) is between the variance contract and two options, the ATM and 2%-OTM puts. It is worth pointing out that the correlations of puts with the market return are always higher (in absolute value) than the correlations with the variance

return. That is, put returns are explained more by directional moves in the market than by the variance return.

Call options are negatively correlated with the variance contract. The correlations are generally smaller (in absolute value) than those for puts. Calls too have stronger correlations with the market return than with the variance return. The correlation between the variance contract and the ATM straddle is only 0.36. The bottom panels in Figure 6 show two scatter plots: the variance return r_v versus the return of the ATM put, and the variance return r_v versus the return of the ATM straddle.

All in all, the variance return seems to contain quite different information from that contained in option returns. To see this intuitively, consider any option that matures at time T . The option payoff at maturity is *path-independent*, because the payoff is a function of the final value F_T only. In contrast, the payoff of the variance contract is *path-dependent*. In other words, while F_T completely determines payoffs of all puts and calls that mature at time T , one needs the whole history of F_t prior to T in order to determine the payoff of the variance contract. This also means that even a collection of infinitely many options with a continuum of strikes does not span the variance return.

To gain additional insights, we note that from the proof of Proposition 2 the payoff of the realized variance contract can be decomposed as:

$$RV_T^{(3)} = OP_T + DS_T, \quad (16)$$

where

$$DS_T := 2 \sum_{i=1}^n \left(\frac{1}{F_{i-1}} - \frac{1}{F_0} \right) (F_i - F_{i-1})$$

is the profit/loss of the dynamic strategy, which rebalances on dates t_i to maintain $2 \left(\frac{1}{F_i} - \frac{1}{F_0} \right)$ shares of the underlying. In this dynamic trading strategy, the initial position in the underlying is set to zero. At time t_i , the position is negative, if $F_i > F_0$, and is positive, if $F_i < F_0$. The dynamic strategy can be viewed as contrarian: the position in the underlying is reduced (increased) when the price goes up (down). This strategy realizes profits when the underlying frequently bounces up and down while remaining in a narrow range of F_0 . On the other hand, the strategy incurs losses when the price trends in either direction. In contrast, the payoff of the portfolio of options OP_T is the higher the farther the underlying moves in either direction from F_0 , with the payoff being zero when $F_T = F_0$. This suggests that the two components OP_T and DS_T should be negatively correlated.

In view of (16), we can decompose the variance return as

$$r_v = \left(\frac{OP_T}{U_0} - 1 \right) + \frac{DS_T}{U_0} = r_{op} + r_{ds},$$

where r_{op} is the return on the static option portfolio, and r_{ds} is the normalized profit/loss of the zero-cost dynamic strategy. The two components are contrasted in the following table:

	Mean	SD	Skew.	Kurt.	α	β	Correlations			
							r_v	r_{op}	r_{ds}	r_m
r_v	-26.30	61.99	4.20	28.37	-23.58	-7.22	1.00	0.46	0.42	-0.50
r_{op}	-47.78	70.78	2.89	14.60	-46.10	-4.45	0.46	1.00	-0.61	-0.27
r_{ds}	21.48	69.31	-0.40	7.75	22.53	-2.77	0.42	-0.61	1.00	-0.17

The results can be summarized as follows. First, the average excess return for the variance contract is decomposed into -47.78% from the option portfolio and 21.48% from the dynamic strategy, indicating that large losses from the option portfolio are partially offset by profits from the dynamic strategy. Huge negative returns for the option portfolio are primarily due to considerable loading on very expensive deep OTM puts. Second, the two components r_{op} and r_{ds} have a strong negative correlation (-0.61), which is consistent with the earlier discussion. This fact also helps explain the relatively low standard deviation for r_v and attractive risk-adjusted performance of the variance contract. Third, both r_{op} and r_{ds} have negative correlations with the market return r_m , but those correlations are much lower (in absolute value) than the correlation between r_v and r_m . Forth, the correlation between the variance return r_v and the return on the static option portfolio r_{op} is not particularly high (0.46), again confirming the intuition that a portfolio of regular options with different strikes does not span the variance return very well.

4 Conclusion

This paper develops a new approach for valuing payoffs tied to the variance. The paper deviates from the existing literature by focusing on the empirically relevant realized variance, as opposed to the unobservable integrated variance. We argue that this distinction is critical for hedging actual variance instruments, such as OTC variance swaps and the CBOE variance futures. In realistic situations, the discretely-sampled realized variance differs considerably from the continuously-sampled integrated variance and the existing approach produces unacceptably large hedging errors.

Our approach is completely model-free. It allows for jumps in the underlying price process and applies to any sampling partition. The approach is useful for several reasons. It identifies the exact replication strategy for the realized variance and shows that hedging errors are very sensitive to even slightest deviations from the optimal timing. The approach allows one to recreate historical prices of the variance contract and to study its risk characteristics. In particular, CBOE can rely on this information to determine the proper margin requirement for its futures, while traders and regulators can use it to assess the extent of the counterparty risk present in OTC variance swaps. We also show how to extend our approach to value the generalized variance contracts, for which the variance is accumulated according to a general weight function.

In the empirical application, we study the economic properties of the variance contract on S&P 500 over the long period from 01/1990 to 12/2009. We find that the variance risk is priced, its risk-premium is negative and very large in economic terms. The variance risk premium cannot be explained by known risk factors and option returns. Trading strategies which involve selling the variance contract would have resulted in very high profits over the studied period. Overall, the empirical findings present a difficult challenge for the theoretical literature to develop an equilibrium model which could justify the size of the variance risk premium observed empirically.¹⁵

¹⁵Several recent papers study the contribution of jumps or stochastic volatility of volatility to the variance risk premium. See Drechsler and Yaron (2001), Bollerslev, Tauchen, and Zhou (2009), among others.

Appendix

A Proofs

Proof of Proposition 1

We first establish the fact that

$$OP_T = 2 \left(\frac{F_T - F_0}{F_0} - \ln \frac{F_T}{F_0} \right). \quad (17)$$

The proof of (17) follows Carr and Madan (1998) and Bakshi and Madan (2000). Let $g(F)$ be a general function, such that $g''(F)$ is continuous almost everywhere. Then the following representation holds

$$g(F_T) = g(x) + g'(x)(F_T - x) + \int_0^x g''(K)(K - F_T)^+ dK + \int_x^\infty g''(K)(F_T - K)^+ dK. \quad (18)$$

for any $x \geq 0$. In particular, when $x = F_0$, the above equation becomes

$$\begin{aligned} g(F_T) &= g(F_0) + g'(F_0)(F_T - F_0) + \int_0^{F_0} g''(K)(K - F_T)^+ dK + \int_{F_0}^\infty g''(K)(F_T - K)^+ dK \\ &= g(F_0) + g'(F_0)(F_T - F_0) + \int_0^\infty g''(K)M_T(K)dK. \end{aligned} \quad (19)$$

Applying (19) to the function $g(F) = \ln F$, one obtains

$$\ln F_T = \ln F_0 + \frac{1}{F_0}(F_T - F_0) - \int_0^\infty \frac{M_T(K)}{K^2} dK,$$

which proves (17). Next, equation (2) and Ito's lemma imply that $d(\ln F_t) = \frac{dF_t}{F_t} - \frac{1}{2}v_t dt$. Therefore,

$$\begin{aligned} IV_T &= \int_0^T v_t dt = -2 \ln \frac{F_T}{F_0} + 2 \int_0^T \frac{dF_t}{F_t} = 2 \left(\frac{F_T - F_0}{F_0} - \ln \frac{F_T}{F_0} \right) + 2 \int_0^T \left(\frac{1}{F_t} - \frac{1}{F_0} \right) dF_t \\ &= OP_T + 2 \int_0^T \left(\frac{1}{F_t} - \frac{1}{F_0} \right) dF_t. \end{aligned} \quad (20)$$

Since the process F_t is a martingale under the risk-neutral measure, the risk-neutral expectation of the dynamic strategy (ii) is zero. This means that $E_0^*[IV_T] = E_0^*[OP_T] = U_0$. $\square$

Proof of Proposition 2

Definition (7) can be re-arranged as

$$\begin{aligned} RV_T^{(3)} &= -2 \ln \frac{F_T}{F_0} + 2 \sum_{i=1}^n \left(\frac{F_i - F_{i-1}}{F_{i-1}} \right) = 2 \left(\frac{F_T - F_0}{F_0} - \ln \frac{F_T}{F_0} \right) + \\ &2 \sum_{i=1}^n \left(\frac{1}{F_{i-1}} - \frac{1}{F_0} \right) (F_i - F_{i-1}) = OP_T + 2 \sum_{i=1}^n \left(\frac{1}{F_{i-1}} - \frac{1}{F_0} \right) (F_i - F_{i-1}), \end{aligned} \quad (21)$$

where the last equality follows from the proof of Proposition 1. Equation (21) identifies the two components of the replication strategy for the payoff $RV_T^{(3)}$: (i) the path-independent payoff OP_T , and (ii) discrete rebalancing in the underlying performed on dates $t_0, t_1, \dots, t_n$. Since F_t is a martingale under the risk-neutral measure, the market price of the realized variance $RV_T^{(3)}$ is equal to U_0 . $\square$

Proof of Proposition 3

It is straightforward to check that $A(y) - A(x) + B(x)(y - x)$ satisfies AP2 for any functions $A(\cdot)$ and $B(\cdot)$. To prove the converse, we first note that if $H(x, y)$ satisfies AP2, then for all x

$$H(x, x) = 0. \quad (22)$$

The above condition follows immediately by considering a trivial martingale of $X_t = X_s = X_u = x$. Now consider another martingale, whose evolution $X_t \rightarrow X_s \rightarrow X_u$ for $t < s < u$ is given as follows:

$$x \rightarrow \begin{cases} x + v & \rightarrow & x + v + \tilde{w} & \text{Prob} = 1/2 \\ x - v & \rightarrow & x - v & \text{Prob} = 1/2, \end{cases}$$

where $v > 0$ and $\tilde{w}$ is a random variable with $E^*[\tilde{w}] = 0$. In words, at time- t the process starts at x , then at time- s with probability $1/2$ it branches to either $(x + v)$ or $(x - v)$, and finally from node $(x + v)$ it changes to $(x + v + \tilde{w})$, but it remains the same at node $(x - v)$. For AP2 to hold, we must have that

$$\begin{aligned} 0 &= E^* [H(X_t, X_u) - H(X_t, X_s) - H(X_s, X_u)] = E^* \left[\frac{1}{2}H(x, x + v + \tilde{w}) + \frac{1}{2}H(x, x - v) \right. \\ &\quad \left. - \frac{1}{2}H(x, x + v) - \frac{1}{2}H(x, x - v) - \frac{1}{2}H(x + v, x + v + \tilde{w}) + \frac{1}{2}H(x - v, x - v) \right] = \\ &\quad \frac{1}{2}E^* [H(x, x + v + \tilde{w}) - H(x, x + v) - H(x + v, x + v + \tilde{w})], \end{aligned}$$

where $E^*[\cdot]$ denotes $E_t^*[\cdot]$. Taking the limit as $v \rightarrow 0$, we obtain

$$E^* [H_2(x, x + \tilde{w}) - H_2(x, x) - H_1(x, x + \tilde{w}) - H_2(x, x + \tilde{w})] = 0,$$

where the subscripts denote the partial derivatives. From (22), $H_1(x, x) = -H_2(x, x)$, implying that

$$E^* [H_1(x, x + \tilde{w})] = H_1(x, x).$$

If we let $h(y) := H_1(x, y)$, then the last equation can be rewritten as $E^* [h(x + \tilde{w})] = h(x)$ and it must hold for any zero-mean random variable $\tilde{w}$. This means that

$$\int_{-\infty}^{\infty} p(w)h(x + w)dw = h(x), \quad \text{for any positive function } p(w) \text{ such that} \quad (23)$$

$$\int_{-\infty}^{\infty} p(w)dw = 1, \quad \text{and} \quad \int_{-\infty}^{\infty} wp(w)dw = 0. \quad (24)$$

The Lagrangian of the system (23)-(24) is $\mathcal{L}(w) = h(x + w) - \lambda_1 - \lambda_2 w$, where λ_1 and λ_2 can depend on x , but not w . The Lagrangian must be zero, implying that $H_1(x, y) = C_1(x) + C_2(x)y$ and

$$H(x, y) = A(y) + B(x)y + D(x)$$

for some functions C_1 , C_2 , A , B , and D . The last equation can be re-written as

$$H(x, y) = A(y) - A(x) + B(x)(y - x) + [D(x) + A(x) + B(x)x].$$

The sum in the square brackets only depends on x and must be zero in view of (22). Therefore,

$$H(x, y) = A(y) - A(x) + B(x)(y - x).$$

□

Proof of Proposition 4

The payoff that we want to replicate can be re-arranged as

$$\begin{aligned}
\sum_{i=1}^n H(F_{i-1}, F_i) &= A(F_T) - A(F_0) + \sum_{i=1}^n B(F_{i-1})(F_i - F_{i-1}) \\
&= A(F_T) - A(F_0) - A'(F_0)(F_T - F_0) + \sum_{i=1}^n (A'(F_0) + B(F_{i-1}))(F_i - F_{i-1}) \\
&= \int_0^\infty A''(K) M_T(K) dK + \sum_{i=1}^n (A'(F_0) + B(F_{i-1}))(F_i - F_{i-1}), \tag{25}
\end{aligned}$$

where the last equality follows from equation (19). Equation (25) identifies the two components of the replication strategy: (i) the path-independent payoff, and (ii) discrete rebalancing in the underlying performed on dates $t_0, t_1, \dots, t_n$. Finally, since F_t is a martingale under the risk-neutral measure, equation (25) implies that time-0 price of the replicated payoff is

$$E_0^* \left[\sum_{i=1}^n H(F_{i-1}, F_i) \right] = \int_0^\infty A''(K) M_0(K) dK dK.$$

□

B Continuous-Time Limit

This appendix derives the continuous-time limit for the three definitions of the realized variance. Suppose that under the objective measure the price follows a general jump-diffusion process:

$$\frac{dF_t}{F_{t-}} = \mu_t dt + \sqrt{v_t} dB_t + \xi_t dJ_t,$$

where J_t is a jump indicator taking the value one if there is a jump at time t and zero otherwise. The jump size ξ_t is non-zero only when $J_t = 1$. Furthermore, the drift μ_t , the variance v_t , and the jump size ξ_t are stochastic processes adapted to the filtration $(\mathcal{F})_{t \in [0, T]}$ and may evolve randomly over time. Finally, for simplicity, we assume that the price jump process is of finite activity.

In this setting, when Δt tends to zero, the realized variances in (5)-(7) converge to the following three version of the quadratic variation:

$$\begin{aligned}
RV_T^{(1)} &\rightarrow QV_T^{(1)} = \int_0^T (d \ln F_t)^2 = \int_0^T v_t dt + \sum_{0 \leq s \leq T} (\ln(1 + \xi_s))^2, \\
RV_T^{(2)} &\rightarrow QV_T^{(2)} = \int_0^T \left(\frac{dF_t}{F_t} \right)^2 = \int_0^T v_t dt + \sum_{0 \leq s \leq T} \xi_s^2, \\
RV_T^{(3)} &\rightarrow QV_T^{(3)} = \int_0^T 2 \left(\frac{dF_t}{F_t} - d \ln F_t \right) = \int_0^T v_t dt + \sum_{0 \leq s \leq T} 2(\xi_s - \ln(1 + \xi_s)),
\end{aligned}$$

where F_s denotes the price before jump ξ_s . The three quadratic variations $QV_T^{(1)}$, $QV_T^{(2)}$, and $QV_T^{(3)}$ differ only with regard to how they treat jumps. When there are no jumps, they all coincide with IV_T . This confirms that Proposition 1 is a special case of Proposition 2 when there are no jumps and $\Delta t \rightarrow 0$.

Similarly, in the continuous-time limit, the generalized variances in (8)-(10) converge to the following

three version of the generalized quadratic variation:

$$\begin{aligned} RV_T^{(1)}(g) &\rightarrow QV_T^{(1)}(g) = \int_0^T g''(F_t) F_t^2 (d \ln F_t)^2 = IV_T(g) + JV_T^{(1)}(g), \\ RV_T^{(2)}(g) &\rightarrow QV_T^{(2)}(g) = \int_0^T g''(F_t) F_t^2 \left(\frac{dF_t}{F_t} \right)^2 = IV_T(g) + JV_T^{(2)}(g), \\ RV_T^{(3)}(g) &\rightarrow QV_T^{(0)}(g) = \int_0^T 2(dg(F_t) - g'(F_t)dF_t) = IV_T(g) + JV_T^{(0)}(g), \end{aligned}$$

where

$$IV_T(g) := \int_0^T g''(F_t) F_t^2 v_t dt,$$

is the generalized integrated variance and

$$\begin{aligned} JV_T^{(1)}(g) &:= \sum_{0 \leq s \leq T} g''(F_s) F_s^2 [\ln(1 + \kappa_s)]^2, \\ JV_T^{(2)}(g) &:= \sum_{0 \leq s \leq T} g''(F_s) F_s^2 \kappa_s^2, \\ JV_T^{(3)}(g) &:= \sum_{0 \leq s \leq T} G(F_s, F_s(1 + \kappa_s)) \end{aligned}$$

are the generalized jump variances. Again, when there are no jumps, the three quadratic variations $QV_T^{(1)}(g)$, $QV_T^{(2)}(g)$, and $QV_T^{(3)}(g)$ all coincide with $IV_T(g)$.

C Data

C.1 CME Futures and Options

The S&P 500 futures have four different maturity months from the March quarterly cycle. The contract size is \$250 times S&P 500 futures price (before November 1997, the contract size was \$500 times futures price). On any trading day, the CME futures options are available for six maturity months: four months from the March quarterly cycle and two additional nearby months ("serial" options). The CME options expire on a third Friday of a contract month. The quarterly options expire at the market open, while the serial options expire at the market close. For the serial options, we measure time to maturity as the number of calendar days between the trade date and the expiration date. For the quarterly options, we use the number of calendar days remaining less one. The option contract size is one S&P 500 futures. The minimum price movement is 0.05. The strikes are multiples of 5 for near-term months and multiples of 25 for far months. If at any time the S&P 500 futures contract trades through the highest or lowest strike available, additional strikes are usually introduced.

C.2 Preparing Dataset

1. For both options and futures we use settlement prices. Settlement prices (as opposed to closing prices) do not suffer from nonsynchronous/stale trading of options and the bid-ask spreads. CME calculates settlement prices simultaneously for all options, based on their last bid and ask prices.
2. In the dataset, we match all puts and calls by trading date t , maturity T , and strike. For each pair (t, T) , we drop very low (high) strikes for which put (call) price is less than 0.1. To convert spot prices to forward prices, we approximate the risk-free rate r_f over $[t, T]$ by the rate of Treasury bills.
3. Because the CME options are American type, their prices $P_t^A(K)$ and $C_t^A(K)$ could be slightly higher than prices of the corresponding European options $P_t(K)$ and $C_t(K)$. The difference, however, is very small for short maturities that we focus on. This is particularly true for OTM and ATM options.

To infer prices of European options $P_t(K)$ and $C_t(K)$, we proceed as follows. First, we discard all ITM options. That is, we use put prices for $K/F_t \leq 1.00$ and call prices for $K/F_t \geq 1.00$. Prices of OTM and ATM options are both more reliable and less affected by the early exercise feature. Second, we correct American option prices $P_t^A(K)$ and $C_t^A(K)$ for the value of the early exercise feature by using Barone-Adesi and Whaley (1987) approximation.¹⁶ Third, we compute prices of ITM options through the put-call parity relationship: $P_t(K) + F_t = C_t(K) + K$.

4. We check option prices for violations of the no-arbitrage restrictions. To preclude arbitrage opportunities, call and put prices must be monotonic and convex functions of the strike. In particular, the call pricing function $C_t(K)$ must satisfy

$$(a) \quad C_t(K) \geq (F_t - K)^+, \quad (b) \quad -1 \leq C'_t(K) \leq 0, \quad (c) \quad C''_t(K) \geq 0.$$

When restrictions (a)-(c) are violated, we enforce them by running the so-called *Constrained Convex Regression* (CCR), introduced in Bondarenko (2000). Intuitively, CCR searches for the smallest (in the sense of least squares) perturbation of option prices that restores the no-arbitrage restrictions. The procedure can also identify possible recording errors or typos.

5. For each pair (t, T) , we estimate RND using the method of *Positive Convolution Approximation*, as described below. Armed with RND, we obtain the put and call pricing functions. The market price of the variance contract is then computed using the relationships in (4).

C.3 Computation of U_0

To estimate the market value of the variance contract U_0 , we use Positive Convolution Approximation (PCA), developed in Bondarenko (2000, 2003a). The method allows one to infer the conditional RND $\hat{h}_0(F_T)$ through the relationship in (1). The method directly addresses the important limitations of option data that (a) options are only traded for a discrete set of strikes, as opposed to a continuum of strikes, (b) very low and very high strikes are unavailable, and (c) option prices are recorded with substantial measurement errors, which arise from non-synchronous trading, price discreteness, and the bid-ask bounce. The PCA method is fully nonparametric, always produces arbitrage-free estimators, and controls against overfitting while allowing for small samples. In addition to the estimate for RND, the method also produces the estimates for the call and put pricing functions $\hat{C}_0(K)$ and $\hat{P}_0(K)$ for a continuum of strikes. Thus, we obtain an estimate $\hat{U}_0$ from (4) by integrating $\hat{C}_0(K)$ and $\hat{P}_0(K)$ over the strike K with the weight function $1/K^2$.

We estimate $U_0 = U(0, T)$ over *calendar* months. For example, for month of June, dates 0 and T in the holding period $[0, T]$ correspond to the last business days in May and June, respectively. However, the S&P 500 options expire not at the end, but roughly in the middle of calendar months. Therefore, we obtain $U_0 = U(0, T)$ by linear interpolation between the two neighboring option maturities T_1 and T_2 which straddle the maturity of the variance contract T . Formally, for $T_1 \leq T \leq T_2$, we compute

$$\frac{U(0, T)}{T} = \frac{T_2 - T}{T_2 - T_1} \cdot \frac{U(0, T_1)}{T_1} + \frac{T - T_1}{T_2 - T_1} \cdot \frac{U(0, T_2)}{T_2}, \quad (26)$$

where $U(0, T_1)$ and $U(0, T_2)$ can be directly estimated from traded options via the relationship in (4). Options with maturities T_1 and T_2 , which correspond to approximately 15 and 45 calendar days, are among most actively traded contracts.

C.4 Computation of Option Returns

We compute returns on *standardized* puts and calls, which at the beginning of each calendar month have fixed moneyness $k = 0.96, 0.98, 1.00, 1.02$, and 1.04 , and which mature at the end of the month. Similar to the variance contract, standardized options are constructed via interpolation between the two closest available maturities T_1 and T_2 ($T_1 < T < T_2$). At the beginning of the month, we estimate RNDs for

¹⁶It is important to point out that this correction is always substantially smaller than typical bid-ask spreads. In particular, the correction generally does not exceed 0.2% of an option price.

T_1 and T_2 , compute Black-Scholes implied volatilities for the five fixed levels of moneyness k , via linear interpolation obtain the corresponding implied volatilities for one-month maturity T , and then convert the implied volatilities into option prices.

D Performance Measures

Sharpe Ratio: This is the most familiar measure of risk-adjusted performance.

$$SH := \frac{E[r_i]}{\sqrt{Var(r_i)}}.$$

Like standard deviation, the Sharpe ratio is inappropriate for non-normally distributed data sets.

Stutzer Index: This measure differs from the Sharpe ratio in that it does not assume that returns are normally distributed. Instead, it takes into account the shape of the distribution of returns. In particular, the Stutzer index penalizes negative skewness and high kurtosis. Where the distribution is normal, the Stutzer index and the Sharpe ratio are identical. For long portfolios ($E(r_i) \geq 0$), the Stutzer index is defined as

$$ST_i := \sqrt{2I_i}, \quad I_i = \max_{\theta} \{ -\ln E[e^{\theta r_i}] \}.$$

For short portfolios ($E(r_i) < 0$), we compute the Stutzer index as minus the index of the corresponding long portfolio.

Sortino Ratio: This measure differs from the Sharpe ratio in that it does not penalize for large upward returns. The Sortino ratio is typically defined in terms of a minimum acceptable return τ for portfolio's return r_i and it measures risk in terms of the corresponding downside deviation.

$$SO := \frac{E[r_i - \tau]}{\sqrt{E[(r_i - \tau)_-^2]}},$$

where $x_- := \min(x, 0)$ and τ is the minimum acceptable return. We set τ to zero taking into account that r_i is already in excess of the risk-free rate. When $E(r_i) < 0$, we assume that portfolio is sold short.

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Table 1: **Parametric Models**

BS	$\mu = 0.11$	$\sigma = 0.15$			
JD	$\mu = 0.11$	$\lambda = 2$	$m_J = -0.03$	$\sigma_J = 0.04$	
SV	$\mu = 0.11$	$\kappa = 5.8$	$\theta = 0.023$	$\sigma_v = 0.35$	$\rho = -0.4$

Notes: This table summarizes parameters for three models: Black-Scholes (BS), Merton jump-diffusion (JD), and Heston stochastic volatility (SV). Under the objective measure, the models are specified as

$$BS : \quad \frac{dF_t}{F_t} = \mu dt + \sigma dB_t, \quad (27)$$

$$JD : \quad \frac{dF_t}{F_{t-}} = \mu dt + \sigma dB_t + (\xi_t dN_t - \lambda \mu_J dt), \quad (28)$$

$$SV : \quad \frac{dF_t}{F_t} = \mu dt + \sqrt{v_t} dB_t, \quad dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t. \quad (29)$$

In the above, B_t and W_t are standard Brownian motion, $E[dB_t dW_t] = \rho dt$, N_t is a Poisson process with constant intensity λ and log-normal jump ξ_t , where $\ln(1 + \xi_t) \sim N[m_J, \sigma_J^2]$ and $\mu_J = e^{m_J + 0.5\sigma_J^2} - 1$. The parameters for the SV model correspond closely to those estimated in Eraker, Johannes, and Polson (2003).

Table 2: **Daily-sampled Realized Variance under Continuous Replication**

	$RV_T^{(1)}$	$RV_T^{(2)}$	$RV_T^{(3)}$
BS	0.392	0.392	0.392
JD	0.382	0.383	0.382
SV	0.403	0.403	0.403

Notes: This table reports hedging errors (HEs) for the three definitions of the realized variance under the assumption that the discretely-sampled payoff is (incorrectly) replicated using the continuous strategy of Proposition 1. The target payoff RV_T has maturity $T = 1$ month and is sampled over equidistant partition with $\Delta t = 1$ day. The parameters for BS, JD, and SV models are set as in Table 1. Hedging errors are computed using Monte-Carlo simulations with 100,000 repetitions, where we assume that one year consists of 252 trading days and one day is equal to 7 trading hours. HE is defined as $HE = \sqrt{E[e^2]}$, where $e = \frac{RV'_T}{RV_T} - 1$.

Table 3: **Hedging Errors****Panel A: Different $\Delta t'$**

		1 min	10 min	1 hour	1/2 day	1 day	2 days	5 days
BS Model	$RV_T^{(1)}$	0.391	0.387	0.359	0.266	0.0024	0.287	0.576
	$RV_T^{(2)}$	0.391	0.387	0.359	0.267	0.0047	0.288	0.583
	$RV_T^{(3)}$	0.391	0.387	0.359	0.266	0	0.287	0.577
JD Model	$RV_T^{(1)}$	0.382	0.377	0.350	0.260	0.0055	0.282	0.565
	$RV_T^{(2)}$	0.381	0.376	0.349	0.260	0.0106	0.283	0.570
	$RV_T^{(3)}$	0.382	0.377	0.350	0.260	0	0.282	0.566
SV Model	$RV_T^{(1)}$	0.407	0.402	0.371	0.275	0.0025	0.295	0.591
	$RV_T^{(2)}$	0.407	0.402	0.371	0.276	0.0050	0.296	0.599
	$RV_T^{(3)}$	0.407	0.402	0.371	0.275	0	0.295	0.593
S&P 500	$RV_T^{(1)}$	0.393	0.325	0.278	0.251	0.0030	0.269	0.466
	$RV_T^{(2)}$	0.393	0.325	0.278	0.251	0.0060	0.270	0.467
	$RV_T^{(3)}$	0.393	0.325	0.278	0.251	0	0.269	0.466

Panel B: Different δt

		0	1 min	5 min	15 min	1 hour
BS Model	$RV_T^{(1)}$	0.0024	0.031	0.070	0.120	0.234
	$RV_T^{(2)}$	0.0047	0.032	0.070	0.121	0.235
	$RV_T^{(3)}$	0	0.031	0.070	0.120	0.234
JD Model	$RV_T^{(1)}$	0.0055	0.031	0.068	0.119	0.231
	$RV_T^{(2)}$	0.0106	0.032	0.069	0.119	0.231
	$RV_T^{(3)}$	0	0.031	0.068	0.118	0.230
SV Model	$RV_T^{(1)}$	0.0025	0.032	0.071	0.125	0.245
	$RV_T^{(2)}$	0.0050	0.032	0.071	0.125	0.245
	$RV_T^{(3)}$	0	0.032	0.071	0.125	0.245
S&P 500	$RV_T^{(1)}$	0.0030	0.049	0.075	0.114	0.253
	$RV_T^{(2)}$	0.0060	0.050	0.076	0.115	0.254
	$RV_T^{(3)}$	0	0.049	0.075	0.114	0.253

Notes: The hedging errors (HEs) are reported for the three definitions of the realized variance and for different $\Delta t'$ (Panel A) or different δt (Panel B), where $\Delta t'$ measures how frequently the underlying is actually rebalanced and δt measures deviations from the optimal times. The target payoff RV_T has maturity $T = 1$ month and is sampled over equidistant partition with $\Delta t = 1$ day. The parameters for BS, JD, and SV models are set as in Table 1. HEs are computed using Monte-Carlo simulations with 100,000 repetitions, where one year consists of 252 trading days and one day is equal to 7 trading hours. For the S&P 500 futures, HEs are computed using high-frequency data over the period from 01/1990 to 12/2009. The relative hedging error is computed as $HE = \sqrt{E[e^2]}$, where $e = \frac{RV'_T}{RV_T} - 1$.

Table 4: Summary Statistics for Monthly Returns

Panel A: Return Distribution									
	Min.	1%	5%	10%	Med.	90%	95%	99%	Max.
VC	-83.42	-81.18	-74.34	-70.24	-40.07	18.61	69.74	199.54	462.83
SP500	-17.25	-11.46	-7.69	-5.38	0.88	5.67	7.06	9.05	10.75
MNP	-398.08	-166.32	-59.92	-26.82	33.22	65.94	78.47	103.78	124.05
MVP	-72.68	-30.73	-10.35	-4.39	6.66	11.32	12.83	15.38	16.94

Panel B: Risk Characteristics									
	Mean	SD	Skew.	Kurt.	α	β	SH	ST	SO
VC	-26.30	61.99	4.20	28.37	-23.58	-7.22	-0.42	-0.34	-0.54
SP500	0.38	4.33	-0.67	4.28	0.00	1.00	0.09	0.09	0.12
MNP	23.58	53.51	-3.30	22.48	23.58	0.00	0.44	0.36	0.60
MVP	4.24	9.44	-3.94	27.37	4.08	0.42	0.45	0.36	0.59

Notes: Statistics are reported for monthly excess returns of the variance contract (VC), the S&P 500 futures, the market-neutral portfolio (MNP), and the mean-variance portfolio (MVP). Returns are expressed in percentage term. MNP is short the variance contract and hedged with S&P 500 futures to have zero correlation with the market return r_m ; its return $r_{mn} = -r_v - 7.22r_m$. MVP is the mean-variance optimal portfolio of the variance contract and S&P 500 futures; its return $r_{mv} = -0.17r_v - 0.83r_m$. Panel A shows minimum, median, maximum, and 1%, 5%, 10%, 90%, 95%, and 99% percentiles. Panel B shows mean, standard deviation (SD), skewness, kurtosis, α and β coefficients (with respect to S&P 500), Sharpe ratio (SH), Stutzer Index (ST), and Sortino ratio (SO). The sample period is 01/1990–12/2009.

Table 5: OLS Regressions for Variance Return

	Const	SP500	SMB	HML	UMD	ATMP	OTMP	ATMC	OTMC	$\bar{R}^2$
Reg 1	-23.58 (-6.79)	-7.22 (-9.02)								0.25
Reg 2	-22.28 (-6.52)	-7.29 (-9.14)	-3.98 (-3.89)	-1.61 (-1.46)						0.29
Reg 3	-20.77 (-6.04)	-8.06 (-9.51)	-4.02 (-3.97)	-2.25 (-2.01)	-1.68 (-2.49)					0.31
Reg 4	-13.78 (-3.60)	-4.41 (-1.97)				0.11 (1.69)	0.10 (2.05)	0.06 (0.94)	0.03 (1.18)	0.35
Reg 5	-11.24 (-2.94)	-5.87 (-2.50)	-3.13 (-3.18)	-2.31 (-2.15)	-1.75 (-2.63)	0.06 (0.95)	0.12 (2.54)	0.05 (0.84)	0.03 (1.05)	0.39

Notes: This table reports results of OLS regressions of the monthly variance return. The regressors include the return on S&P 500 futures, the Fama-French SMB and HML factors, the momentum factor UMD, returns of ATM put ($k = 1.00$), OTM put ($k = 0.96$), ATM call ($k = 1.00$), and OTM call ($k = 1.04$). All returns are in excess of risk-free rate and expressed in percentage term; t -statistics are reported in parentheses. The last column reports the adjusted R^2 .

Table 6: **Risk Characteristics for Monthly Returns of S&P 500 Options**

Puts											
k	Mean	SD	Skew.	Kurt.	α	β	SH	ST	SO	ρ_m	ρ_v
0.96	-60.21	129.27	3.97	20.14	-52.71	-19.87	-0.47	-0.38	-0.56	-0.67	0.55
0.98	-44.29	130.56	2.74	10.63	-35.71	-22.71	-0.34	-0.30	-0.42	-0.75	0.57
1.00	-28.67	120.72	1.85	5.97	-19.88	-23.29	-0.24	-0.22	-0.30	-0.84	0.57
1.02	-18.83	99.45	1.34	4.34	-10.95	-20.89	-0.19	-0.18	-0.25	-0.91	0.55
1.04	-12.06	78.82	0.99	3.77	-5.46	-17.48	-0.15	-0.15	-0.21	-0.96	0.53

Calls											
k	Mean	SD	Skew.	Kurt.	α	β	SH	ST	SO	ρ_m	ρ_v
0.96	-0.97	66.77	0.14	2.28	-6.36	14.30	-0.01	-0.01	-0.02	0.93	-0.39
0.98	-3.22	84.82	0.48	2.44	-9.64	17.03	-0.04	-0.04	-0.05	0.87	-0.33
1.00	-6.33	111.02	1.07	3.49	-13.84	19.91	-0.06	-0.06	-0.07	0.78	-0.25
1.02	-17.80	147.31	2.08	7.63	-26.08	21.92	-0.12	-0.12	-0.14	0.65	-0.16
1.04	-48.84	170.44	5.27	37.06	-56.02	19.02	-0.29	-0.24	-0.32	0.48	-0.06

ATM Straddle											
k	Mean	SD	Skew.	Kurt.	α	β	SH	ST	SO	ρ_m	ρ_v
1.00	-17.50	58.05	0.83	3.65	-16.86	-1.69	-0.30	-0.29	-0.48	-0.13	0.36

Notes: Statistics are reported for excess returns of 1-month options on the S&P 500 futures: puts and calls with different moneyness k and ATM straddles. Statistics include mean, standard deviation (SD), skewness, kurtosis, α and β coefficients (with respect to S&P 500), Sharpe ratio (SH), Stutzer Index (ST), Sortino ratio (SO), and correlations with the market and variance contract. Returns are expressed in monthly percentage term.

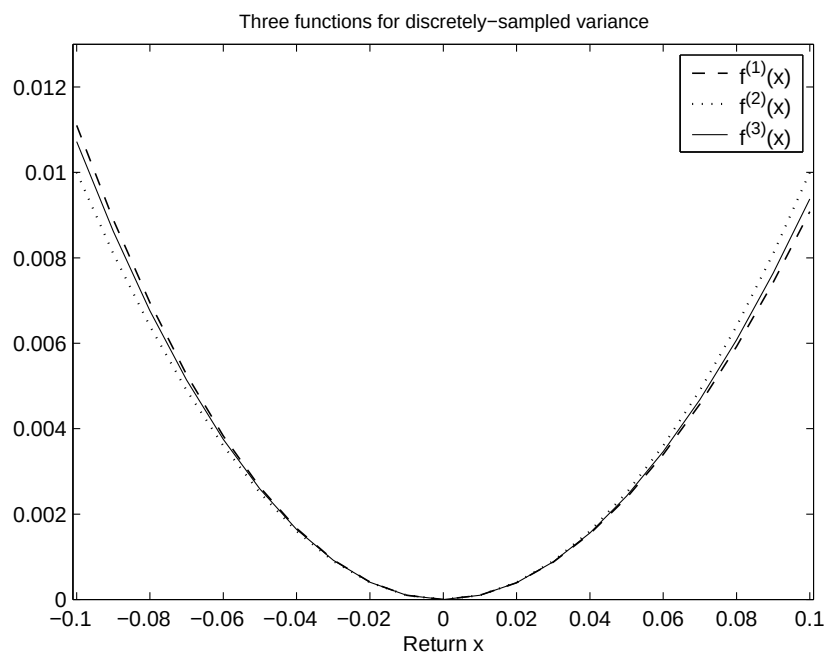


Figure 1: This figure shows behavior of functions $f^{(1)}(x) = [\ln(1+x)]^2$, $f^{(2)}(x) = x^2$, and $f^{(3)}(x) = 2(x - \ln(1+x))$ used in the definitions of the discretely-sampled variance $RV_T^{(1)}$, $RV_T^{(2)}$, and $RV_T^{(3)}$.

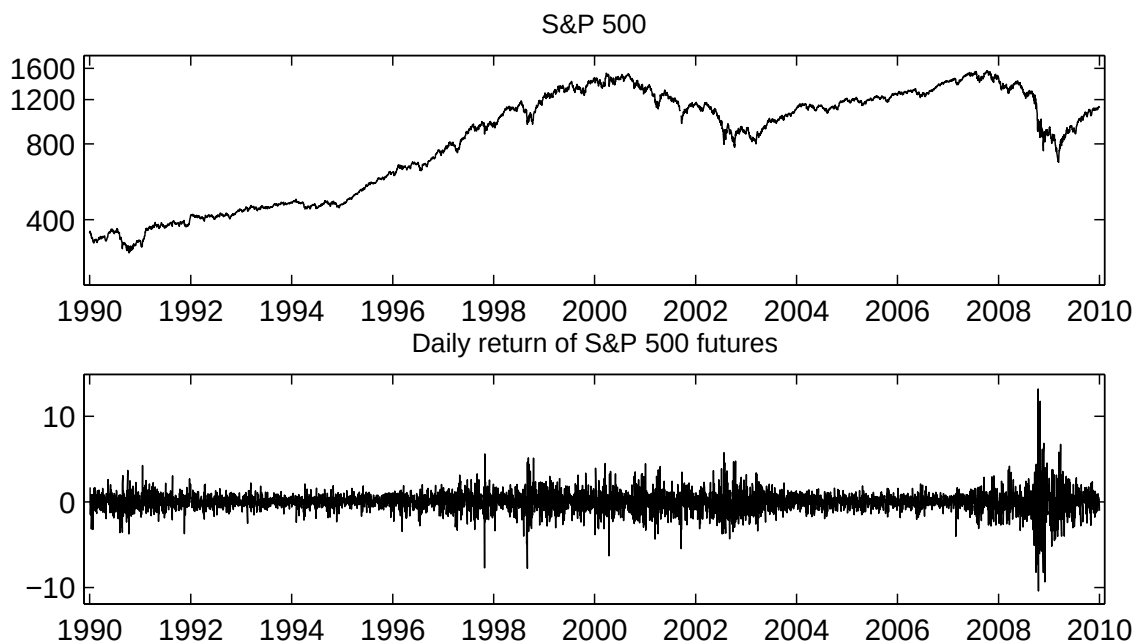


Figure 2: This figure plots the level of the S&P 500 and daily return of the S&P 500 futures. The sample period is from 01/1990 to 12/2009.

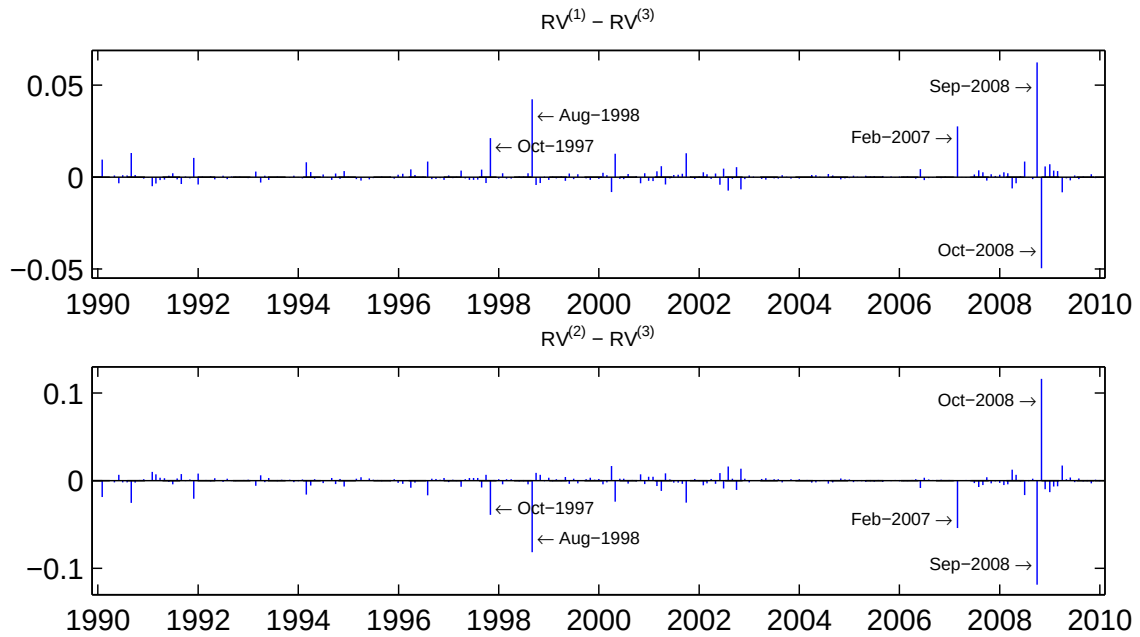


Figure 3: This figure plots the deviation of the alternative variance payoffs $RV_T^{(1)}$ and $RV_T^{(2)}$ from $RV_T^{(3)}$. The holding period is one month. The deviations are normalized by U_0 .

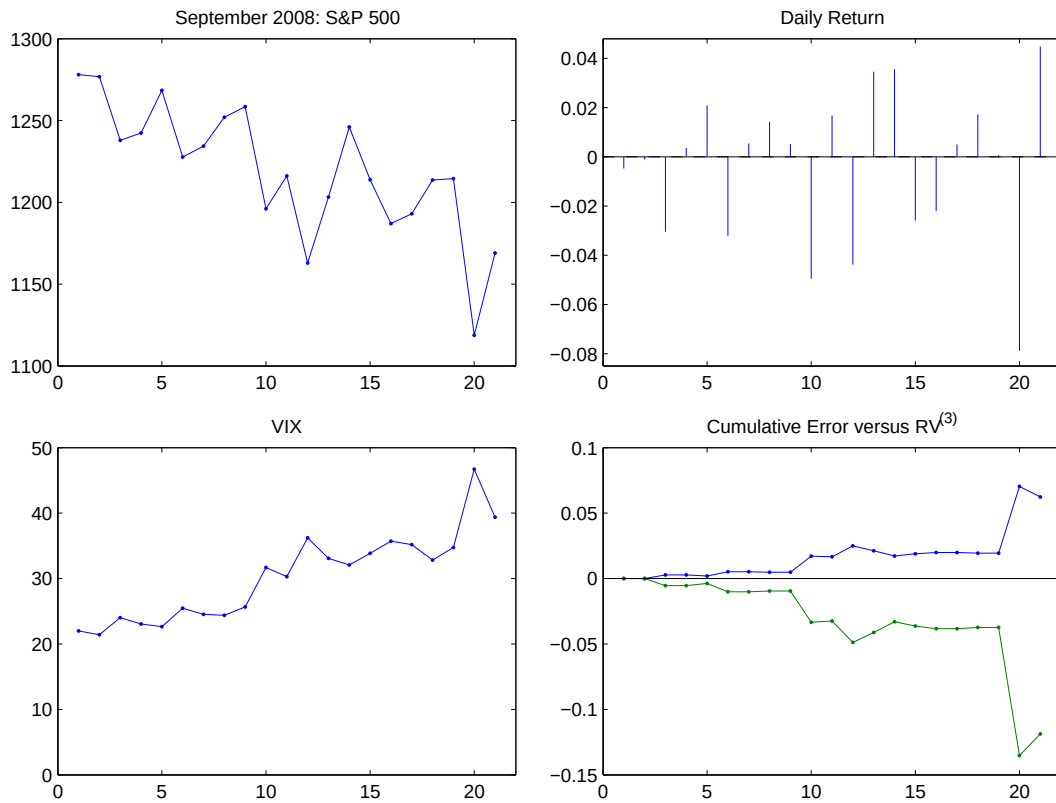


Figure 4: This figure illustrates September 2008. The first three panels show S&P 500 futures, its daily return, and VIX. The bottom right panel shows the cumulative deviation the alternative variance payoffs $RV_T^{(1)}$ (higher line) and $RV_T^{(2)}$ (lower line) from $RV_T^{(3)}$. The deviations are normalized by U_0 .

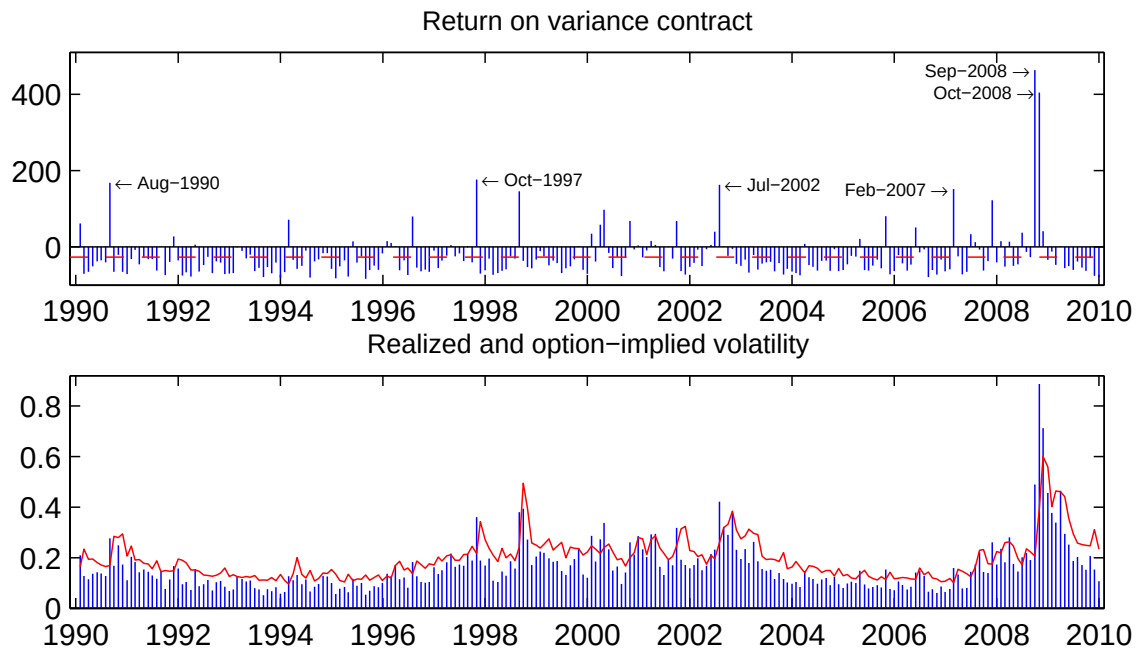


Figure 5: The top panel plots the monthly variance return. Its sample mean is shown with the dashed line. The bottom panel plots the square root of the market price of the variance contract (the red line) and the realized variance of S&P 500 futures (the vertical bars). The holding period is one month. Both series are annualized.

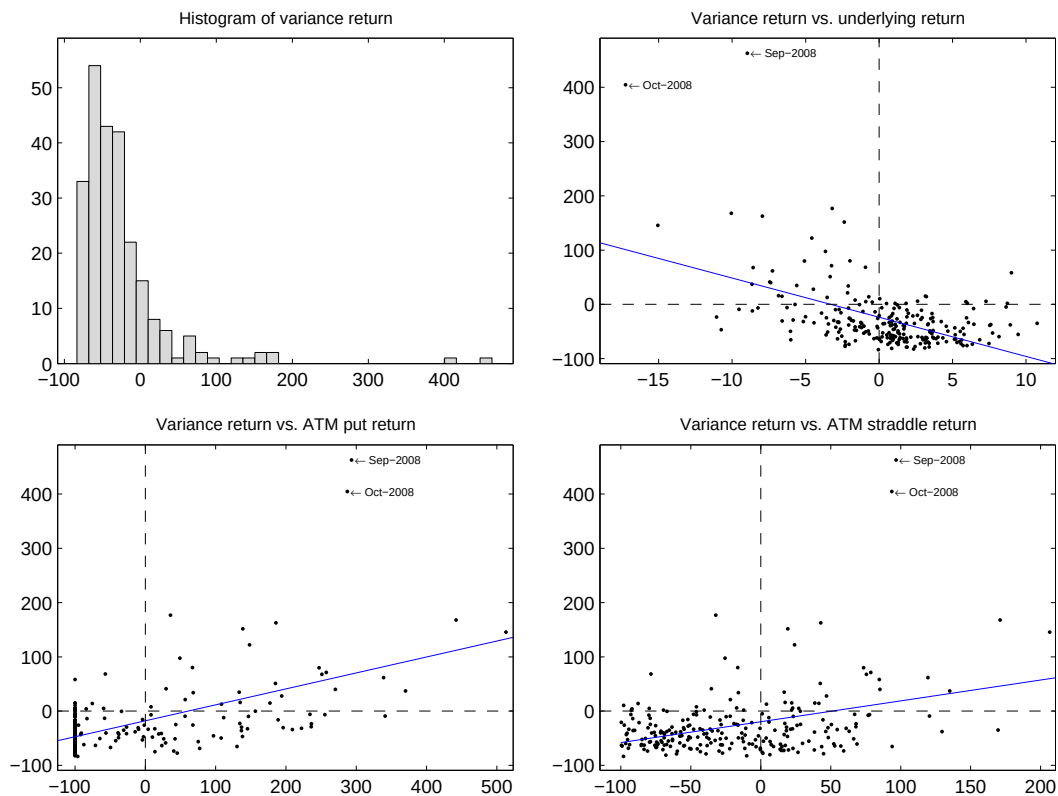


Figure 6: The top left panel shows the histogram of the variance return. The three other panels are scatter plots of the variance return versus the return on the underlying S&P 500 futures, the ATM put, and the ATM straddle. Also shown are corresponding OLS regression lines. All returns are monthly.